



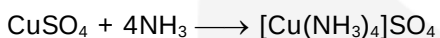
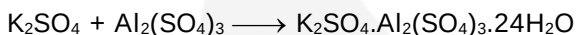
Introduction

- (a) The concept of co-ordination compounds arise from the complex formation tendency of transition elements.
- (b) These compounds play a vital role in our lives, as chlorophyll of plants, vitamin B₁₂ and hemoglobin of animal blood are the co-ordination compounds of Mg, Co and Fe respectively.
- (c) The co-ordination compounds play important role in analytical chemistry, polymerization reaction, metallurgy and refining of metals photography, water purification etc.
- (d) Co-ordination compounds also find many applications in electroplating, textile dyeing and medicinal chemistry.

Addition Compounds

When solutions containing two or more salts in simple molecular proportion are evaporated, crystals of new compound separate out.

Example:



These addition compounds can be divided into two classes:

(A) Double Salts:

Those which lose their identity in solution

In solutions these compounds break down into simpler ions. Such addition compounds which lose their identity in solutions are called double salts.

Example:

Potash alum $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ when dissolved in water breaks down into K^+ , SO_4^{2-} , Al^{+3} ions of therefore is an example of double salt.

(B) Coordination Compounds:

Those which retain their identity in solution.

In aqueous solution, these addition compounds do not furnish all simple ions but instead give more complex ions having complicated structure.

Example:

Potassium ferrocyanide $\text{K}_4[\text{Fe}(\text{CN})_6]$ does not furnish simple K^+ , Fe^{2+} and CN^- ions but gives K^+ ions and complex ferrocyanide ions, $[\text{Fe}(\text{CN})_6]^{4-}$. These types of compounds are called co-ordination compound.

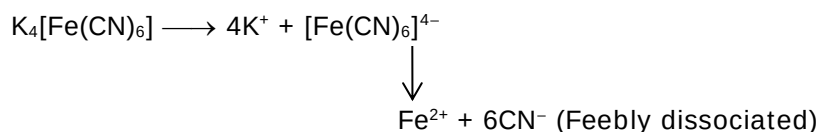
Complex compounds or co-ordination compounds.



On the basis of stability of complex ion, complex ions are further divided as follows –

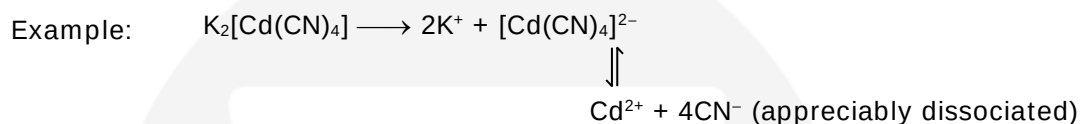
- (i) Perfect complexes: The compounds in which complex ion is fairly stable and further dissociation is not possible in solution state.

Example:



The ferrocyanide ion $[\text{Fe}(\text{CN})_6]^{4-}$ is so insignificantly dissociated that it can be considered as practically undissociated and does not give the qualitative test of Fe^{2+} or CN^- ions.

- (ii) Imperfect complexes: Those complexes in which complex ion is less stable and is reversibly dissociated to give enough simple ions and thus respond to their usual qualitative test.



Note: Homoleptic Complexes: (Metal cation or atom surrounded by same type of ligand)

Example: $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$, $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

Heteroleptic Complexes: (Metal cation or atom surrounded by different type of ligand)

Example: $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{PtCl}_2(\text{NH}_3)_2]$

Illustration:

1. Study the following double salt and write the type of ions which can be tested in their aqueous solution.

Salt

Ions can be tested

- (a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (b) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (c) $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (d) $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (e) $\text{Al}_2(\text{OH})_3\text{PO}_4 \cdot \text{H}_2\text{O}$

Solution:

Salt

Ions can be tested

- (a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ K^+ , Al^{3+} , SO_4^{2-}
 (b) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ NH_4^+ , Al^{3+} , SO_4^{2-}
 (c) $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ K^+ , Cr^{3+} , SO_4^{2-}
 (d) $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ Fe^{2+} , Al^{3+} , SO_4^{2-}
 (e) $\text{Al}_2(\text{OH})_3\text{PO}_4 \cdot \text{H}_2\text{O}$ Al^{3+} , OH^- , PO_4^{3-}



2. Predict which among the following properties given below belong to double salt and co-ordination compounds.
- (a) The blue colored solution prepared by Cu^{+2} (aq.) and $\text{NH}_3(\text{NH}_4^+ \text{ aq.})$ does not show the presence of Cu^{+2} (.....)
 - (b) Compounds which are stable in the solid state but break up into its constituents in aqueous solution (.....)
 - (c) Aqueous solution of carnallite (.....)
 - (d) The compounds in which central metal ion form dative bonds with species surrounding it. (.....)
 - (e) Mohrs salt (.....)

Solution:

- (a) Co-ordination (b) Double salt (c) Double salt
- (d) Co-ordination (e) Double salt

Definitions Of Terms Used In Co-Ordination Compounds

- (a) Co-ordination or complex compound: Co-ordination compounds are those molecular compounds which retain their identity even when dissolved in water or any other solvent and their properties are completely different from those of the constituent ions.
- (b) Central ion: The cation to which one or more neutral molecules / ions are attached is called the atom / ion. Since, the central ion acts as an acceptor and thus has to accommodate electron pairs donated by the donor atoms of neutral molecules or ions, it must have empty orbitals of appropriate energy.
- (c) Complex ion: A complex ion may be defined as an electrically charged radical which is formed by the combination of a simple cation with one/more neutral molecules or one/more simple.
- (d) Co-ordination number: The total number of co-ordinate covalent bond is formed by central metal in complex is called the co-ordination number of the central metal ion .

Some common co-ordination numbers of important metals are as given below.

Metal	Coordination Number	Metal	Coordination Number
Cu^+	2, 4	Ni^{2+}	4, 6
Ag^+	2	Fe^{2+}	4, 6
Au^+	2, 4	Fe^{3+}	6
Hg_2^{2+}	2	Co^{2+}	4, 6
Cu^{2+}	4, 6	Co^{3+}	6
		Al^{3+}	6
Pt^{2+}	4	Sc^{3+}	6
Pd^{2+}	4	Pt^{4+}	6
Mg^{2+}	6	Pd^{4+}	6



Example:

Coordination number of the central metal ions in

(i) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is four

(ii) $[\text{Fe}(\text{EDTA})]^-$ is six

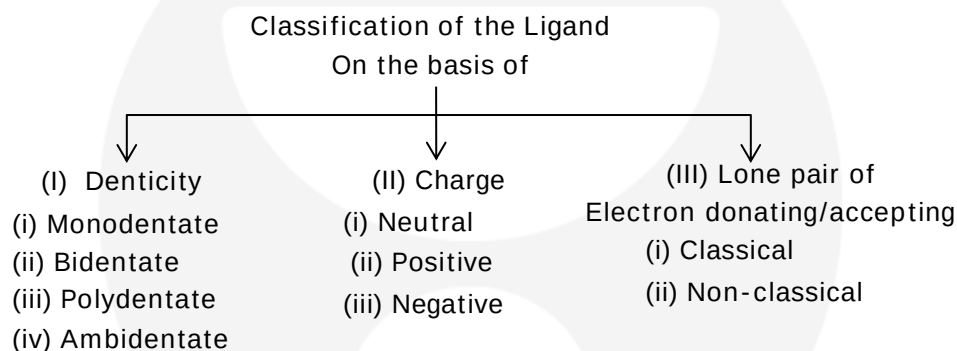
(e) Co-ordination sphere: The part of the complex enclosed in square bracket is known as co-ordination sphere. It is actually combination of central metal and ligands.

(f) Ligands

(i) The ions or neutral molecules which combine with central metal ion to form complex are called ligands.

(ii) They act as electron pair donor (i.e. Lewis bases) though certain ligands also accept electron from central metal and such ligands are known as π - acid ligands.

Classification Of Ligands



(A) Based on charge

(i) Neutral ligands: H_2O , NO , CO , C_6H_6 etc.

(ii) Positive ligands: NO^+ , N_2H_5^+

(iii) Negative ligands: Cl^- , NO_2^- , CN^- , OH^-

(B) Based on Denticity

The number of electron pairs donated to central metal by ligands is known as denticity.

Depending on number of electron pairs donated, these are classified in following categories.

(a) Unidentate/monodentate ligands: Ligands which donate one pair of electron to the central metal are called unidentate ligands.

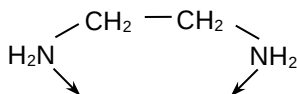
X^- , CN^- , NO_2^- , NH_3 , Pyridine, OH^- , NO_3^- , H_2O , SO_3^{2-} , CO , NO , OH^- , O^{2-} , $(\text{C}_6\text{H}_5)_3\text{P}$ etc.

(here: $\text{X}^- \rightarrow \text{F}^-$, Cl^- , Br^- , I^-)

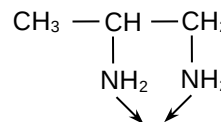


- (b) Bidentate ligands: Ligands which have two donor atoms and have the ability to link with central metal ion at two positions are called bidentate ligands.

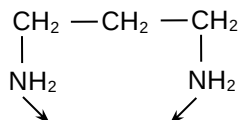
(i) en: ethylenediamine



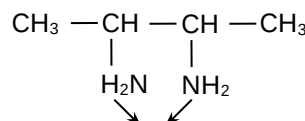
(ii) pn: propylenediamine



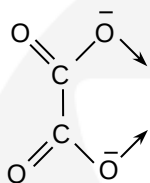
(iii) tn: trimethylenediamine



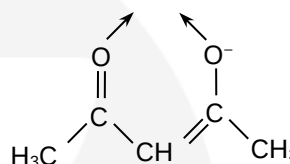
(iv) bn: butylenediamine



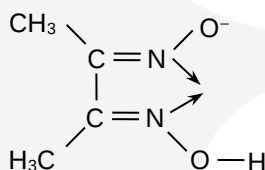
(v) ox^{2-} : oxalate



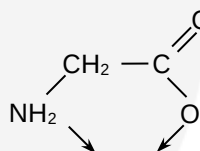
(vi) acac⁻: acetylacetonate



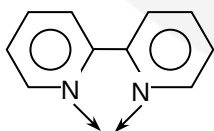
(vii) dmgl⁻: dimethylglyoximate



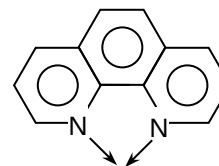
(viii) gly⁻: glycinate



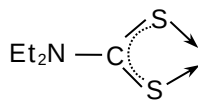
(ix) dipy: dipyridyl



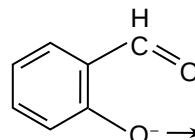
(x) o-phen: ortho-phenanthroline



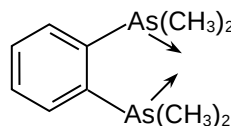
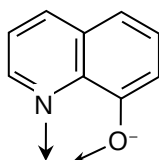
(xi) N, N'- Diethylthiocarbamate ion



(xii) Salicylaldehyde anion



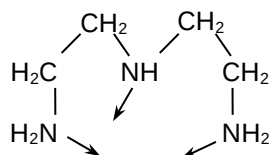
(xiii) 8-Hydroxyquinolinol ion (oxine) (xiv) o-Phenylenebisdimethylarsine (diarsine)



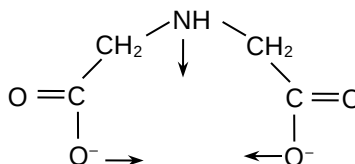


- (c) Tridentate ligands: The ligands which donate three pairs of electrons to the central metal are called tridentate ligands. For example,

(i) dien: diethylenetriamine

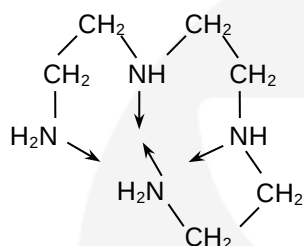


(ii) imda²⁻: iminodiacetate

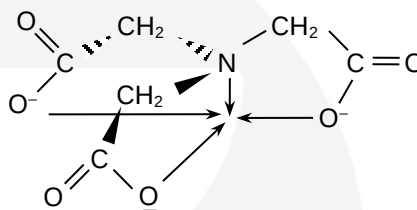


- (d) Tetradentate ligands: Those ligands which can donate four electron pair to the central metal are known as tetradentate ligands. For example,

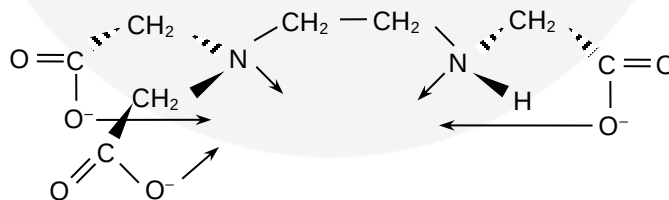
(i) trien: triethylenetetraamine



(ii) NTA³⁻: nitrilotriacetate

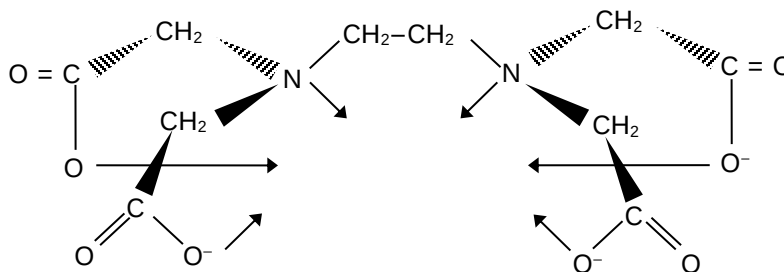


- (e) Pentadentate ligands: Those ligands which can donate five electron pairs to the central metal are known as pentadentate ligands. For example, EDTA³⁻: ethylenediamine triacetate



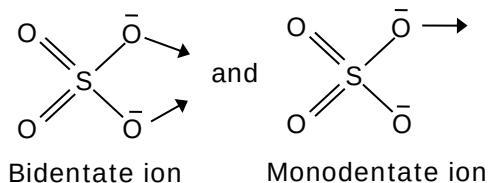
- (f) Hexadentate ligands: Those ligands which can donate six electron pairs to the central metal are known as hexadentate ligands. For example,

EDTA⁴⁻: ethylenediamine tetracetate





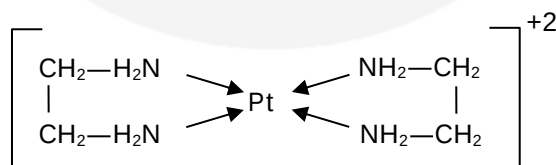
Note: (i) Flexidentate: A ligand which shows variable denticity is called a flexidentate ligand. For example,



- (ii) $[\text{Co}(\text{NH}_3)_4\text{CO}_3]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{Br}$
 CO_3^{2-} is bidentate CO_3^{2-} is monodentate
 Example:
 $\text{CO}_3^{2-}, \text{SO}_3^{2-}$ etc.

(g) Chelating ligands

- (i) Polydentate ligands whose structures permit the attachment of two or more donor sides to the same metal ion simultaneously, thus closing one or more rings are called chelating ligands and the compounds formed are known as chelate compounds.
- (ii) A chelate may be defined as a ring structure formed by the combination of a polydentate ligand having two or more donor atoms with a metal ion forming part of the ring.
- (iii) The process of formation of chelates is called chelation.
- (iv) Chelate complexes are more stable than ordinary complexes in which the ligand is a monodentate
- (v) This increased stability of the compound due to chelation is called the chelate effect
- (vi) In the complex ion given below, 5 membered rings are formed. So all these are called chelate complex



- (vii) Generally the chelate complexes with 5 or 6 membered rings are more stable.
 - (viii) Out of these, 5 membered rings are very stable when they involve saturated ligands.
- On the other hand 6-membered ring structures acquire maximum stability when they involve unsaturated ligands containing conjugate double bond. This is due to the resonance effects involving metal d-orbitals and ligand p-orbital electrons.



(h) Ambidentate ligands

Ligands which can be bonded through two different atoms present in it are called ambidentate ligands. Examples of such ligands are the NO_2^- and SCN^- ions. NO_2^- , The negative ion can coordinate through either the nitrogen or the oxygen atoms to a central metal atom/ion. Similarly, SCN^- ion can coordinate through the sulphur or nitrogen atom. Such possibilities give rise to linkage isomerism in coordination compounds. There are certain ligands which have two or more donor atoms but during formation of complexes only one donor atom is attached to metal ion. Such ligands are called ambidentate ligands such as CN^- , CNS^-



Example:

CN^- can coordinate through either the nitrogen or the carbon atom to central metal ion.

Illustration:

Match the Column.

A		B	
(i)	Ligand contains one donor site	(a)	hexadentate
(ii)	Ligand contains two donor site	(b)	tridentate
(iii)	Ligand contain three donor site	(c)	tetradentate
(iv)	Ligand contain four donor site	(d)	bidentate
(v)	Ligand contain six donor site	(e)	unidentate

Ans. (i) - (e); (ii) - (d); (iii) - (b); (iv) - (c); (v) - (a)

(C) Based upon bonding interaction between the ligand and the central atom.

(a) Classical or simple ligand: These ligand only donate the lone pair of electrons to the central atom.

eg.: O^{2-} , OH^- , F^- etc.

(b) Non classical or π -acid or π -acceptor ligand: These ligand not only donate the lone pair of electrons to central metal but also accept the electron cloud from central atom

eg.: CO , CN^- , NO^+ , PF_3 , PR_3 etc.



Table
Important Ligands forming Complexes

Type of Ligand	Name of Ligand	Symbol	Structure showing only linkage sites (by $\rightarrow$ to metal ions)
Unidentate	Water	aqua	
	Ammonia	ammine	
	pyridine	py	
Bidentate	Ethylenediamine	en	
	Glycinate anion	gly	
	Acetylacetonate anion	acac	
	Oxalate anion	ox	
	Dimethylglyoximate anion	DMG [⊖]	
	2, 2'Dipyridyl	dipy	
	1,10 Phenanthroline	phen	



Type of Ligand	Name of Ligand	Symbol	Structure showing only linkage sites (by $\rightarrow$ to metal ions)
Bidentate	N,N'-Diethylthiocarbamate ion		
	Salicylaldehyde anion		
	8-Hydroxyquinolinol ion	oxine	
	o-Phenylenebisdimethylarsine	diarsine	
Tridentate	Diethylene triamine	dien	
	Iminodiacetate anion	imda ²⁻	
	2,2',2'' Terpyridine	terpy	



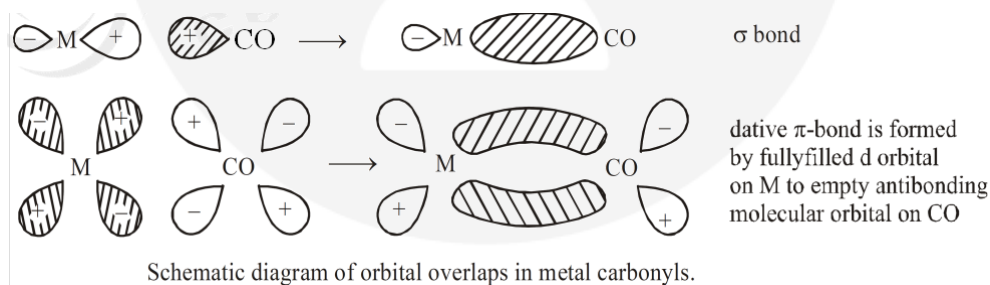
Type of Ligand	Name of Ligand	Symbol	Structure showing only linkage sites (by $\rightarrow$ to metal ions)
Tetradentate	Triethylene tetraamine	trien	
	Nitrilotriacetato	nta ⁻³	
Pentadentate	Ethylenediamine triacetate anion	EDTA ⁻³	
Hexadentate	Ethylenediamine tetraacetate anion	EDTA ⁻⁴	



- Q.1 Which of the following is monodentate ligands?
 (A) en (B) dien (C) DMG (D) OH⁻
- Q.2 Some salts although containing two different metallic elements give test for one of them in solution. Such salts are:
 (A) Complex salt (B) Double salt (C) Normal salt (D) None of these
- Q.3 Which of the following is copper(I) compound?
 (A) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (B) $[\text{Cu}(\text{CN})_4]^{3-}$ (C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (D) All of these
- Q.4 Diethylenetriamine is:
 (A) chelating agent (B) tridentate neutral molecule
 (C) tridentate monoanion (D) both (A) and (B)
- Q.5 Some salts although containing two different metallic elements give test for one of them in solution. Such salts are:
 (A) double salts (B) complex salts (C) normal salts (D) none of these

Bonding In Metal Carbonyl

Example:



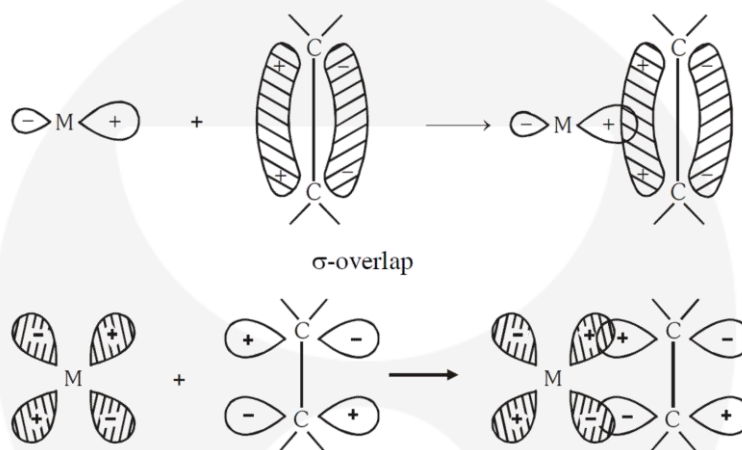
- (a) The metal-carbon bond in metal carbonyls may be represented as the donation of an electron pair from carbon to vacant orbital of metal & form σ bond ($\text{M} \leftarrow \text{CO}$).
- (b) A second bond is formed by back bonding sometimes called dative π -bonding. This bond formation take place from sideways overlap of a filled orbital on the metal with the empty antibonding π^*p_y/π^*p_z (if x-axis is molecular axis) molecular orbital of the carbon monoxide, thus forming a $\text{M} \longrightarrow \text{CO}$ bond ($d\pi-p\pi$ back bond).
- (c) The filling or partial filling, of the antibonding orbital on CO reduces the bond order of C–O bond from the triple bond in CO towards a double bond. This shown by the increase in C–O bond length from 1.128 Å in CO to about 1.15 Å in many carbonyls.



- (d) Since CO accept the back donated electron from the metal atom in to its vacant π^* orbital, CO is called π -acid or π -acceptor ligand or π -bonding ligand.
- (e) Other π -acid ligands are CN^- , RCN , NO^+ , PF_3 , PPh_3 etc.

Bonding in π -bonded organo metallic compound

Like zeise's salt $\text{K}[\text{PtCl}_3(\pi\text{-C}_2\text{H}_4)]$, the bonding of alkenes to a transition metal to form complexes has two type of bonds. First, the π -electron density of the alkene overlaps with a σ -type vacant orbital of the metal atom. Second is the π back bond formed by the flow of electron density from a filled d-orbital on the metal into the vacant π^* - antibonding molecular orbital on the alkene molecule as shown below:



IUPAC Nomenclature of Coordination Compounds

The main rules of naming of complexes are –

- (a) Like simple salts, the positive part of the coordination compound is named first.
Example:
 $\text{K}_4[\text{Fe}(\text{CN})_6]$ the naming of this complex starts with potassium.
 $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ the naming of this complex starts with name of complex ion.
- (b) Naming of coordination sphere:- The names of ligands along with their numerical prefixes (to represent their number) are written first, followed by the name of central metal.
- (c) The ligands can be neutral, anionic or cationic.
- (i) The neutral ligands are named as the molecule Ex. $(\text{C}_5\text{H}_5\text{N})$ pyridine, $(\text{C}_6\text{H}_5)_3\text{P}$ Triphenyl phosphine, $(\text{H}_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2)$ ethylene diamine.
The neutral ligands which are not named as the molecule are CO (carbonyl), NO (nitrosyl), H_2O (Aqua), NH_3 (ammine).



- (ii) Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	Name as ligand	Symbol	Name as ligand
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo/Peroxido
CN^-	Cyano/Cyanido		
O^{2-}	Oxo/Oxido	S^{2-}	Sulphido
OH^-	Hydroxo/Hydroxido	NH^{2-}	Imido
H^-	Hydrido	NH_2^-	Amido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

Symbol	Name as ligand	Symbol	Name as ligand
CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$\text{C}_2\text{O}_4^{2-}$	Oxalato	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	ONO^-	(bonded through oxygen) nitrito
NO_3^-	Nitrato	NO_2^-	(bonded through nitrogen) nitro/nitrito-N
$\text{S}_2\text{O}_3^{2-}$	Thiosulphato		

- (iii) Positive ligands naming ends in 'ium' $\text{NH}_2\text{—NH}_3^+$ Hydrazinium, NO_2^+ nitronium, NO^+ nitrosonium/nitrosylium.

- (d) If ligands are present more than once, then their number is indicated by prefixes like di, tri, tetra etc.
- (e) If words like di, tri, tetra are already used in the naming of ligand, or if it is polydentate ligand or organic ligand, the prefixes bis-, tris-, tetrakis-, pentakis- etc. are used to specify their number.

Example: $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$: Dichlorobis(ethylenediamine)platinum(IV) chloride.

- (f) When more than one type of ligands are present in the complex, then the ligands are named in the alphabetical order.



- (g) After naming of ligands the central metal ion is to be named followed by its oxidation state in Roman numbers in brackets. (as per IUPAC)

If the complex is neutral or provides a cationic complex ion, then the central metal ion is to be named as it is.

If the complex provides anionic complex ion then the name of central metal ion ends with 'ate'

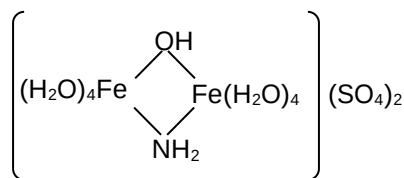
Example: $(\text{NH}_4)_2[\text{CuCl}_4]$: Ammonium tetrachlorocuprate(II)

- (h) After the naming of central metal ion, anion which is in the outer sphere is to be named. The naming of some of the complexes is done as follows – (as per IUPAC)

Complex Compounds		IUPAC Name
(i)	$\text{K}_4[\text{Fe}(\text{CN})_6]$ (anionic complex) So suffix 'ate' is added with metal name	Potassium hexacyanidoferrate(II)
(ii)	$\text{K}_2[\text{PtCl}_6]$	Potassium hexachloridoplatinate(IV)
(iii)	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (Cationic complex) So metal name will be written without any suffix	Hexamminecobalt(III) chloride
(iv)	$[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$	Tetraaquadichloridochromium(III) chloride
(v)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	Diamminetetrachloridoplatinum(IV)
(vi)	$[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (Neutral complex) So no suffix is used with name of metal ion	Triamminetrichloridocobalt(III)
(vii)	$\text{K}_3[\text{Co}(\text{NO}_2)_6]$	Potassium hexanitrocobaltate(III)
(viii)	$\text{Na}_3[\text{Fe}(\text{CN})_5\text{NO}]$	Sodium pentacyanidonitrosoniumferrate(II)
(ix)	$[\text{NiCl}_4]^{-2}$	Tetrachloridonickelate(II) ion
(x)	$[\text{Ru}(\text{NH}_3)_5\text{Cl}]^{+2}$	Pentamminechloridoruthenium(III) ion
(xi)	$[\text{Fe}(\text{en})_3]\text{Cl}_3$	Tris-(ethylenediamine)iron(III) chloride
(xii)	$[\text{Ni}(\text{Gly})_2]$	Bis-(glycinato)nickel(II)

- (i) If a complex ion has two metal atoms then it is termed polynuclear. The ligand which connects the two metal ions is called as Bridging ligand or Bridge group.

A prefix of Greek letter μ , is repeated before the name of each different kind of bridging group.



Tetraaquaairon(III)-μ-amido-μ-hydroxidotetraaquaairon(III) sulphate

Illustration:

Write the systematic names of the following co-ordination entities and compounds:

- (i) $[\text{CoCl}_2(\text{NH}_3)_3]^+$ (ii) $[\text{CrCl}_3(\text{NH}_3)_3]$ (iii) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ (iv) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (v) $[\text{PtCl}(\text{NH}_3)_5]\text{Cl}_3$

Solution:

- (i) triaamminedichloridocobalt(III)
 (ii) triamminetrichloridochromium(III)
 (iii) potassium trioxalatochromate(III)
 (iv) potassium hexacyanidoferrate(II)
 (v) pentaamminechloridoplatinum(IV) chloride

Concept Builders



- Q.6 The formula of the complex tris-(ethylenediamine)cobalt(III) sulphate is
 (A) $[\text{Co}(\text{en})_2\text{SO}_4]$ (B) $[\text{Co}(\text{en})_3\text{SO}_4]$
 (C) $[\text{Co}(\text{en})_3]_2\text{SO}_4$ (D) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$
- Q.7 The IUPAC name of $[\text{Ni}(\text{CO})_4]$ is
 (A) tetracarbonylnickelate(0) (B) tetracarbonylnickelate(II)
 (C) tetracarbonylnickel(0) (D) tetracarbonylnickel(II)
- Q.8 The complex $\text{Hg}[\text{Co}(\text{CNS})_4]$ is correctly named as:
 (A) mercury(II) tetrathiocyanatocobaltate(II)
 (B) mercury(II) cobalttetrasulphocyano(II)
 (C) mercury(II) tetrasulphocyanidecobaltate(II)
 (D) mercury(II) sulphocyanatocobalt(II)
- Q.9 The hypothetical complex triamminediaquachloridocobalt(III) chloride can be represented as
 (A) $[\text{CoCl}(\text{NH}_3)_3(\text{H}_2\text{O})_2]$ (B) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{Cl}_3]$
 (C) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{Cl}]\text{Cl}_2$ (D) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$
- Q.10 The correct IUPAC name of the complex $[\text{Fe}(\pi\text{-C}_5\text{H}_5)_2]$ is
 (A) cyclopentadienyliron (II) (B) bis-(η^5 -cyclopentadienyl)iron(II)
 (C) dicyclopentadiencyferrate (II) (D) ferrocene



Some Important Compounds With Their Common Name

- Switzer Reagent: $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
- **Nessler's Reagent:** $\text{K}_2[\text{HgI}_4] + \text{KOH}$
- Cisplatin: $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
- Brown Ring Complex: $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}^+]\text{SO}_4^-$
- Sodium Nitro Prusside: $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}^+]^-$
- Wilkinson Catalyst: $[\text{Rh}(\text{PPh}_3)_3\text{Cl}]^+$
- **Vaska's Complex:** $[\text{Ir}(\text{Cl})(\text{CO})(\text{PPh}_3)_2]$
- Magnus Green Salt: $[\text{Pt}(\text{NH}_3)_4]^{+2}[\text{PtCl}_4]^{-2}$
- **Collman's Reagent:** $\text{Na}_2[\text{Fe}(\text{CO})_4]^{-2}$
- **Prussian Blue and Turnbull's blue:** Prussian Blue and Turnbull's blue is the same substance made from different reagents.
 $\text{Fe}^{3+} + [\text{Fe}^{2+}(\text{CN})_6]^{4-} \longrightarrow \text{Fe}_4^{3+}[\text{Fe}^{2+}(\text{CN})_6]_3$ (Prussian Blue)
 $\text{Fe}^{2+} + [\text{Fe}^{3+}(\text{CN})_6]^{3-} \longrightarrow \text{Fe}_3^{3+}[\text{Fe}^{2+}(\text{CN})_6]_2$ (Turnbull's Blue)

Formation Of Co-Ordination Compounds

It can be explained by number of theories.

- (A) Werner's co-ordination theory
- (B) Sidgwick theory or Effective Atomic Number Theory (EAN)
- (C) Valence bond theory
- (D) Crystal field theory

(A) Werner's Co-Ordination Theory

Werner's co-ordination theory was the first attempt to explain the bonding in co-ordination compounds. The main postulates of this theory are:

- (a) Metals possesses two types of valencies - Primary valency and secondary valency.
- (b) Primary valencies are ionisable and are exhibited by a metal in the formation of its simple salts such as CoCl_3 , CuSO_4 and AgCl . In these salts the primary valencies of Co, Cu and Ag are 3, 2, 1 respectively. Now-a-days primary valencies are referred to as oxidation state of their metal ion.



- (c) Secondary valencies are non-ionisable and are exhibited by a metal in the formation of its complex ions such as $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and $[\text{Ag}(\text{NH}_3)_2]^+$. In these complex, the secondary valencies of Co^{3+} , Cu^{2+} , Ag^+ are 6, 4 and 2 respectively. These are referred to as co-ordination number (C.N.) of the metal cation.
- (d) Primary linkages (valencies) are satisfied by negative ions while secondary valencies are satisfied by neutral molecules, negative ions.
- (e) Every metal atom or ion has a fixed number of secondary valencies. In other words, the co-ordination number of the metal atom is usually fixed.
- (f) Every metal has tendency to satisfy both its primary and secondary valencies.
- (g) The ligands satisfying secondary valency are always directed towards fixed positions in space about the central metal atom or ion. Thus, the co-ordination compounds have a definite geometry. Werner deduced that in $\text{CoCl}_3 \cdot 5\text{NH}_3$ only two of the three chlorine atoms are ionic and 5 NH_3 and one Cl form co-ordinate bonds to Co^{3+} ion.
- Formula of some cobalt complexes.

Example:

Old		New	No. of Cl^- Ions precipitated	Total No. of ions
(i)	$\text{CoCl}_3 \cdot 6\text{NH}_3$	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	3	4
(ii)	$\text{CoCl}_3 \cdot 5\text{NH}_3$	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	2	3
(iii)	$\text{CoCl}_3 \cdot 4\text{NH}_3$	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$	1	2

Complex	Modern formula	No. of Cl^- Ions precipitated	Total No. of ions
$\text{PtCl}_4 \cdot 6\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$	4	5
$\text{PtCl}_4 \cdot 5\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	3	4
$\text{PtCl}_4 \cdot 4\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$	2	3
$\text{PtCl}_4 \cdot 3\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$	1	2
$\text{PtCl}_4 \cdot 2\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	0	0 (non-electrolyte)



Werner's Representation Of Complexes

(i)	$\text{Fe}(\text{NH}_3)_6\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_6]\text{Cl}_3$ Dotted lines indicate primary valency and continuous lines indicate secondary valency of metal ion.
(ii)	$\text{Fe}(\text{NH}_3)_5\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ In this complex two 'Cl' group act as primary valency and one of the 'Cl' acts as secondary valency also.
(iii)	$\text{Fe}(\text{NH}_3)_4\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ In this complex one 'Cl' group act as primary valency and two of the 'Cl' groups act as secondary valency also.

(B) Sidgwick Theory or Effective Atomic Number (EAN)

Sidgwick proposed effective atomic number theory to explain the stability of the complexes. Total number of electrons on central metal including those transferred from ligands is known as

EAN. The EAN generally coincides with the atomic number of next inert gas except in some cases.

EAN can be calculated by the following relation:

$\text{EAN} = (\text{atomic number of the metal} - \text{oxidation state of central metal}) + \text{number of electrons gained from the donor atoms of the ligands.}$

Example: Effective atomic number of the metal atom in the following:

(a) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ is 33

(b) $\text{K}_4[\text{Fe}(\text{CN})_6]$ is 36

Illustration:

When AgNO_3 is added to a solution of $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, the precipitate of AgCl shows two ionisable chloride ions. This means -

(A) Two chlorine atom satisfy primary valency and one chlorine atom satisfy primary valency as well as secondary valency.

(B) One chlorine atom satisfies primary valency.

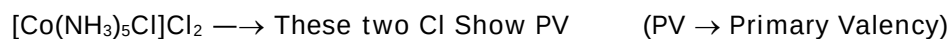
(C) Two chlorine atoms satisfy secondary valency.

(D) Three chlorine atoms satisfy secondary valency.

Ans. (A)



Solution:



These one Cl show PV as well SV (SV $\rightarrow$ Secondary Valency)

Complex	Metal (Oxidation State)	Atomic Number or Metal	Coordination number	Effective atomic number (E-A.N.)
$\text{K}_4[\text{Fe}(\text{CN})_6]$	+2	26	6	$(26-2) + (6 \times 2) = 36$ [Kr]
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$	+2	29	4	$(29-2) + (4 \times 2) = 35$
$[\text{Co}(\text{CH}_3)_6]\text{Cl}_3$	+3	27	6	$(27-3) + (6 \times 2) = 36$ [Kr]
$\text{Ni}(\text{CO})_4$	0	28	4	$(28-0) + (4 \times 2) = 36$ [Kr]
$\text{K}_2[\text{Ni}(\text{CN})_4]$	+2	28	4	$(28-2) + (4 \times 2) = 34$
$\text{K}_2[\text{PtCl}_6]$	+4	78	6	$(78-2) + (6 \times 2) = 86$ [Rn]
$\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$	+3	24	6	$(24-3) + (6 \times 2) = 33$
$\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$	+3	26	6	$(26-3) + (6 \times 2) = 35$
$\text{K}_2[\text{HgI}_4]$	+2	80	4	$(80-2) + (4 \times 2) = 86$ [Rn]
$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	+1	47	2	$(47-1) + (2 \times 2) = 50$
$\text{K}_2[\text{PdCl}_4]$	+2	46	4	$(46-2) + (4 \times 2) = 52$

Concept Builders



- Q.11 Which one of the following complex species does not obey the EAN rule?
 (A) $[\text{Cu}(\text{CN})_4]^{3-}$ (B) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (C) $[\text{Fe}(\text{CN})_6]^{4-}$ (D) $[\text{Ni}(\text{CO})_4]$
- Q.12 EAN of a metal carbonyl $\text{M}(\text{CO})_x$ is 36. If atomic number of metal M is 26, what is the value of x?
 (A) 4 (B) 8 (C) 5 (D) 6
- Q.13 Following Sidgwick's rule of EAN, $\text{Co}(\text{CO})_x$ will be:
 (A) $\text{Co}_2(\text{CO})_4$ (B) $\text{Co}_2(\text{CO})_3$ (C) $\text{Co}_2(\text{CO})_8$ (D) $\text{Co}_2(\text{CO})_{10}$
- Q.14 Coordination number of platinum in $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$ ion is:
 (A) 4 (B) 2 (C) 8 (D) 6
- Q.15 In dodecacarbonyl dicobalt, the oxidation state of Cobalt is:
 (A) 1 (B) 2 (C) 3 (D) Zero



(C) Valence Bond Theory

The main features of this theory are -

- Every metal ion when it forms a complex compound undergoes formation of coordinate covalent bond.
- During this bond formation, the metal ion acts as electron pair acceptor. For this the metal ion provides vacant orbitals.
- The number of vacant orbitals provided is equal to the coordination number of metal ion.
Example: In the formation of $[\text{Fe}(\text{NH}_3)_6]^{3+}$, Fe^{+3} ion provides six vacant orbitals.
In $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Cu^{+2} ion provides four vacant orbitals.
- The metal provides vacant orbitals only after the process of hybridisation, thus vacant hybrid orbitals are provided by the metal ion.
- The vacant hybrid orbitals of metal ion get overlapped by orbitals of ligands containing lone pair of electrons.
- The number of such overlappings is equal to the coordination number of metal ion.
- The empty 'd' orbitals involved in hybridisation may be inner (n-1)d or outer "nd" orbitals and accordingly complexes are called as Inner orbital complexes and outer orbital complexes respectively.
- In certain complexes pairing of electrons takes place in ligand field, resulting in decrease in spin only magnetic moment, such complexes are known as Low spin complexes
- Paramagnetism is represented in the term of spin only magnetic moment.

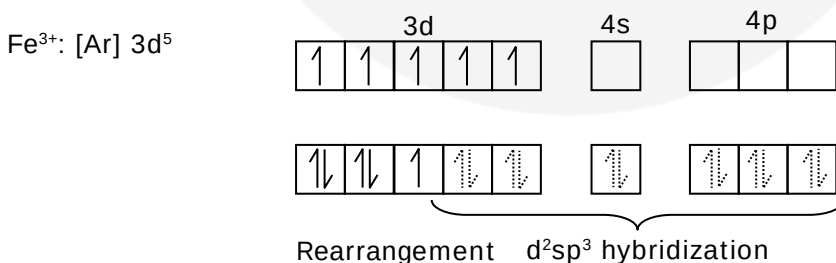
$$\mu = \sqrt{n(n+2)} \text{ B.M.} \quad n = \text{Number of unpaired electrons}$$

Example:

$[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic while $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic.

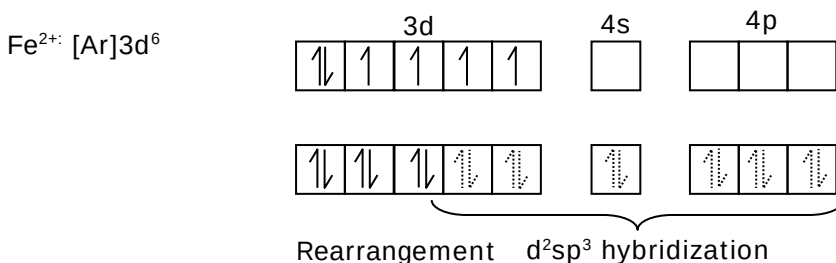
Solution:

$[\text{Fe}(\text{CN})_6]^{3-}$ involves d^2sp^3 hybridization



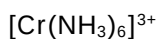
One d-orbital is singly occupied, hence it is weakly paramagnetic in nature.

$[\text{Fe}(\text{CN})_6]^{4-}$ also involves d^2sp^3 hybridization but it has Fe^{2+} ion as central ion.





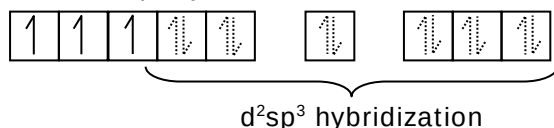
All electrons are paired, hence it is diamagnetic in nature.



(a) Cr: $[\text{Ar}]4s^1 3d^5$



(c) Cr^{3+} in d^2sp^3 hybridized state



(d) This d^2sp^3 hybridisation leads to octahedral geometry. The complex $[\text{Cr}(\text{NH}_3)_6]^{3+}$ will be octahedral in shape

(e) Complex ion has 3 unpaired electrons so it must be paramagnetic.

(f) Other complexes of chromium with similar inner structure are $[\text{Cr}(\text{CN})_6]^{3-}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

Outer orbital complexes

(a) In these complexes s, p as well as d orbitals involved in hybridization, belong to the highest quantum number (n)

(b) Complexes formed by the use of outer nd orbitals will be paramagnetic.

(c) These complexes are called high-spin or spin free complexes

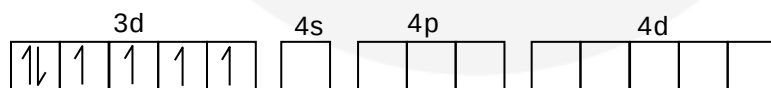
(d) The outer orbital complexes have greater number of unpaired electrons.



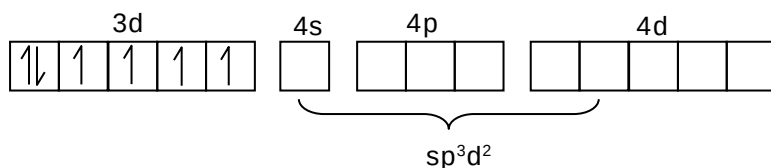
(a) $_{27}\text{Co}$



(b) Co^{3+} ion



(c) Co^{3+} ion in sp^3d^2 hybridised state



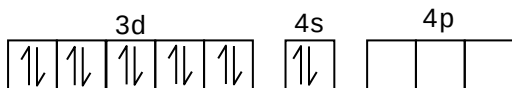
(d) Due to octahedral orientation of six sp^3d^2 hybridised orbitals shape $[\text{CoF}_6]^{3-}$ complex ion is octahedral

(e) Due to presence of four unpaired electrons in 3d orbital $[\text{CoF}_6]^{3-}$ ion has paramagnetic character

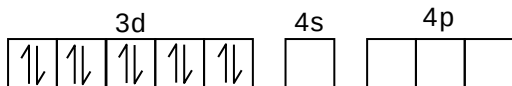
(f) Other examples are $[\text{FeF}_6]^{3-}$, $[\text{Fe}(\text{NH}_3)_6]^{2+}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Cu}(\text{NH}_3)_6]^{2+}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$



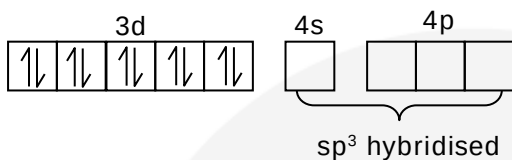
- (a) Configuration of Zn_{30} : $[\text{Ar}]4s^23d^{10}$



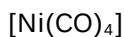
- (b) Electronic configuration of Zn^{3+} : $[\text{Ar}]4s^03d^{10}$



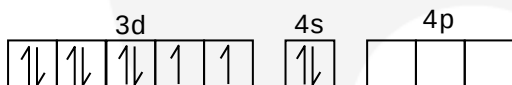
- (c) Zn^{2+} in sp^3 hybridised state



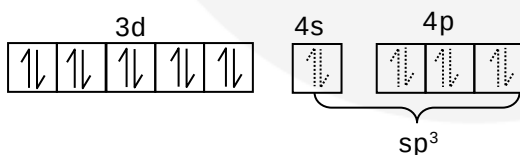
- (d) Since the complex is formed by sp^3 hybridisation, it is tetrahedral
- (e) Since all electrons are paired it is diamagnetic



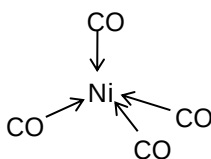
- (a) Ni_{28} : $[\text{Ar}]4s^23d^8$



- (b) Ni after rearrangement: $[\text{Ar}]4s^03d^{10}$



- (c) The empty one 4s and three 4p orbitals mix to form four sp³- hybridised orbitals
- (d) each orbital accepts one electron pair from carbon monoxide molecule forming $[\text{Ni}(\text{CO})_4]$
- (e) The shape of nickel tetra carbonyl is tetrahedral as shown below
- (f) is diamagnetic in nature





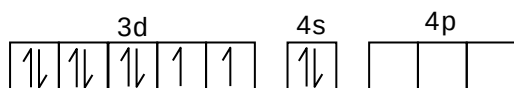
Square planar complex

These are formed due to dsp^2 hybridisation.

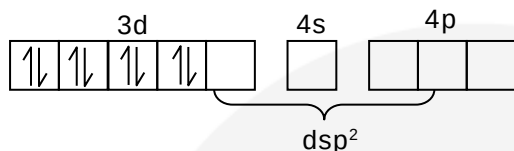
These complexes tend to be formed when the central ion has only one d orbital available in the inner shell



- (a) Electronic configuration of Ni: $[\text{Ar}]4s^23d^8$



- (b) Electronic configuration of Ni^{2+} : $[\text{Ar}]4s^03d^8$



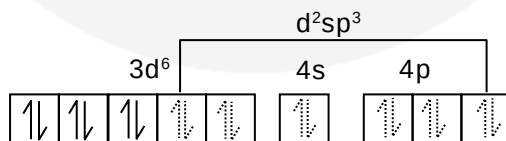
- (c) The rearrangement is due to presence of strong ligand CN^- . The four orbitals make dsp^2 hybridisation
- (d) The shape of resulting complex is square planar
- (e) Due to paired electrons it is diamagnetic.

Illustration:

$[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic while $[\text{FeF}_6]^{4-}$ is strongly paramagnetic. Why?

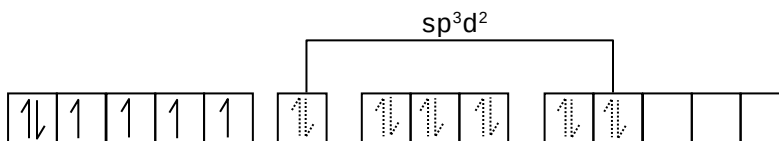
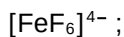
Solution:

CN^- is strong field ligand and so it pairs up the electrons leading to the formation of low spin complex which is diamagnetic.



Octahedral, diamagnetic complex

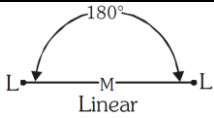
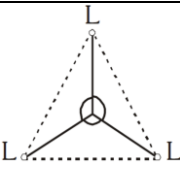
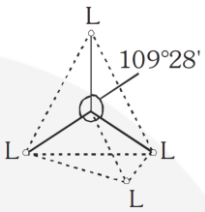
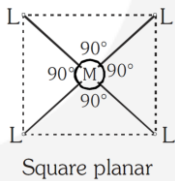
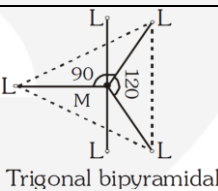
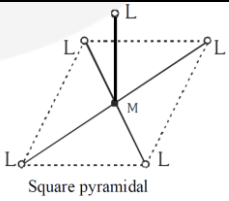
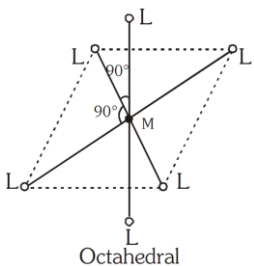
F^- is weak field ligand so it forms high spin paramagnetic complex.



Octahedral, paramagnetic complex



Some Example:

Coordination Number	Hybridised orbitals	Geometrical shape of the complex	Examples of Complex
2	sp		$[\text{Ag}(\text{NH}_3)_2]^+$ $[\text{Ag}(\text{CN})_2]^-$
3	sp^2		$[\text{HgI}_3]^-$
4	sp^3		$[\text{CuCl}_4]^{2-}$ $[\text{ZnCl}_4]^{2-}$ $[\text{FeCl}_4]^-$ $[\text{Ni}(\text{CO})_4]$ $[\text{Zn}(\text{NH}_3)_4]^{+2}$
4	dsp^2 $d = d_{x^2-y^2}$		$[\text{PdCl}_4]^{2-}$ $[\text{Ni}(\text{CN})_4]^{2-}$ $[\text{Pt}(\text{NH}_3)_4]^{+2}$ $[\text{Cu}(\text{NH}_3)_4]^{+2}$ $[\text{PtCl}_4]^{2-}$
5	sp^3d ($d = d_{z^2}$) or dsp^3 ($d = d_{z^2}$)		$[\text{Fe}(\text{CO})_5]$ $[\text{CuCl}_5]^{3-}$
5	sp^3d ($d = d_{x^2-y^2}$) or dsp^3 ($d = d_{x^2-y^2}$)		$[\text{Ni}(\text{CN})_5]^{-3}$
6	d^2sp^3 (inner orbital complex) or sp^3d^2 (outer orbital complex) in both case d-orbitals are d_{z^2} & $d_{x^2-y^2}$.		$[\text{Cr}(\text{NH}_3)_6]^{+3}$ $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$ $[\text{Fe}(\text{CN})_6]^{-3}$ $[\text{Co}(\text{NH}_3)_6]^{+3}$ $[\text{PtCl}_6]^{-2}, [\text{CoF}_6]^{-3}$



Drawback of valence bond theory

- (a) It describes bonding in co-ordination compounds only qualitatively but not account for the relative stabilities for different co-ordination complexes.
- (b) It does not offer any explanation for optical absorption spectra (coloration) of complexes
- (c) It does not describe the detailed magnetic properties of co-ordination compounds.

Concept Builders



- Q.16 What will be spin only magnetic moment of $[\text{Fe}(\text{CN})_2(\text{H}_2\text{O})_4]\text{NO}_3$?
(A) $\sqrt{8}$ B.M. (B) $\sqrt{15}$ B.M. (C) $\sqrt{24}$ B.M. (D) $\sqrt{35}$ B.M.
- Q.17 The complex $[\text{Pt}(\text{NH}_3)_4]^{2+}$ has structure:
(A) square planar (B) tetrahedral (C) pyramidal (D) pentagonal
- Q.18 Which has maximum paramagnetic nature?
(A) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (C) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (D) $[\text{Fe}(\text{CN})_6]^{4-}$
- Q.19 Which is correct in the case of $[\text{NiCl}_4]^{2-}$ complex?
(A) sp^3 hybridisation (B) Paramagnetic and tetrahedral
(C) Two unpaired electrons (D) All are correct
- Q.20 The shape of $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$ is:
(A) Tetrahedral (B) Octahedral (C) Square planar (D) Pyramidal

(D) Crystal Field Theory

The drawbacks of VBT of coordination compounds are, to a considerable extent, removed by the Crystal Field Theory. The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interaction between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d-orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of polar molecules like NH_3 and H_2O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lost. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.



(a) Crystal field splitting in octahedral coordination entities

In an octahedral coordination entity with six ligands surrounding the metal atom/ion, there will be repulsion between the electrons in d orbitals of metal and the electrons (or negative charges) of the ligands. Such repulsion is more when the d orbitals of metal are directed towards the ligand than when it is away from the ligand.

Thus, the $d_{x^2-y^2}$ and d_{z^2} orbitals (axial orbitals) which point towards the axis along the direction of the ligand will experience more repulsion and will be raised in energy ; and the d_{xy} , d_{yz} and d_{zx} orbitals (non-axial) orbitals which are directed between the axis will be lowered in energy relative to the average energy in the spherical crystal field.

Thus, the degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o (the subscript o is for octahedral). Thus, the energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of the three t_{2g} will decrease by $(2/5)\Delta_o$.

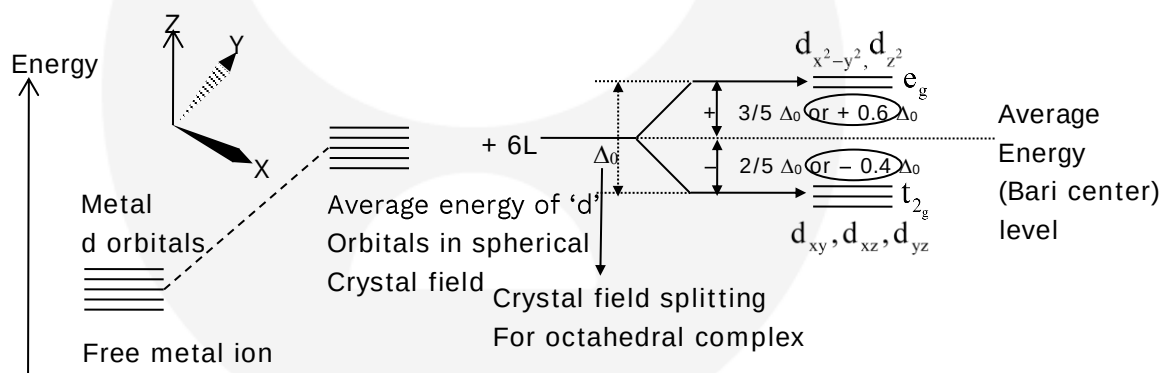
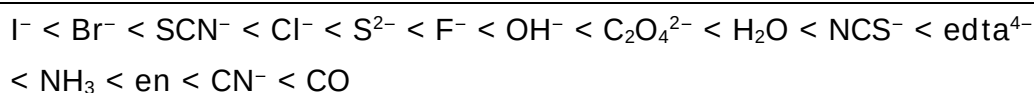


Figure showing crystal field splitting in octahedral complex.

The crystal field splitting, Δ_o , depends upon the fields produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak fields and consequently result in small splitting of d orbitals. In general, ligands can be arranged in a series in the order of increasing field strength as given below:





Note: In SCN^- , S is donating atom and in NCS^- , N is donating atom.

Such a series is termed as spectrochemical series. It is an experimentally determined series based on the absorption of light by complexes with different ligands. For d^4 configuration, the fourth electron will singly occupy e_g orbital (according to Hund's rule) or will undergo pairing in t_{2g} orbital, which of these possibilities occurs, depends on the relative magnitude of the crystal field splitting, Δ_o and the pairing energy, P (P represents the energy required for electron pairing in a single orbital). The two possibilities are:

- (i) If $\Delta_o < P$, the fourth electron enters in one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$.

Ligands for which $\Delta_o < P$ are known as weak field ligands and form high spin complexes.

- (ii) If $\Delta_o > P$, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4 e_g^0$. Ligands which produce this effect are known as strong field ligands and form low spin complexes.

Crystal Field stabilising energy in Octahedral field

Formula: $\text{CFSE} = [-0.4 n_{t_{2g}} + 0.6 n_{e_g}] \Delta_o + xP$.

Where $n_{t_{2g}}$ & n_{e_g} are number of electron(s) in t_{2g} & e_g orbitals respectively and Δ_o crystal field splitting energy for octahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

- (b) Crystal field splitting in tetrahedral coordination entities:

In tetrahedral coordination entity formation, the d orbital splitting is inverted and is smaller as compared to the octahedral field splitting. For the same metal, the same ligands and metal-ligand distances, it can be shown that $\Delta_t = (4/9)\Delta_o$. This may be attributed to the following two reasons.

- (i) There are only four ligands instead of six, so the ligand field is only two thirds the size; as the ligand field splitting is also two thirds the size.
- (ii) The direction of the orbitals does not coincide with the direction of the ligands. This reduces the crystal field splitting by roughly further two thirds.

$$\text{So } \Delta_t = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9} \Delta_o.$$

Consequently, the orbital splitting energies are not sufficiently large for forcing pairing and, therefore, low spin configurations are rarely observed.

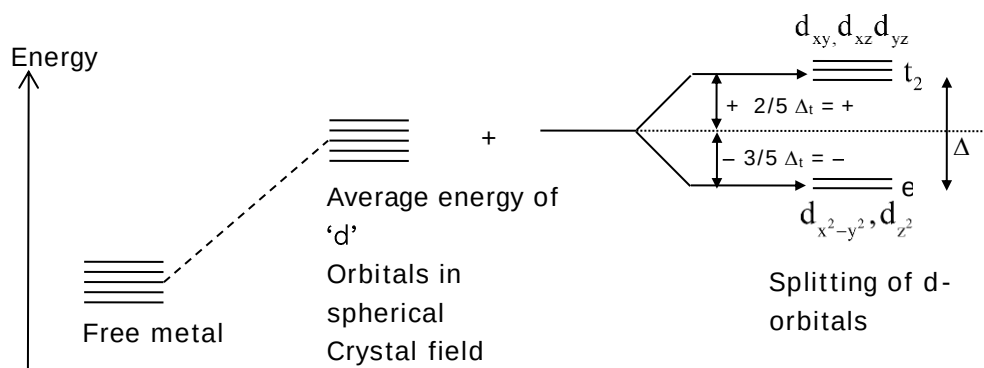


Figure showing crystal field splitting in tetrahedral complex.

Since $\Delta_t < \Delta_o$ crystal field splitting favours the formation of octahedral complexes.

Crystal Field stabilising energy in Tetrahedral field

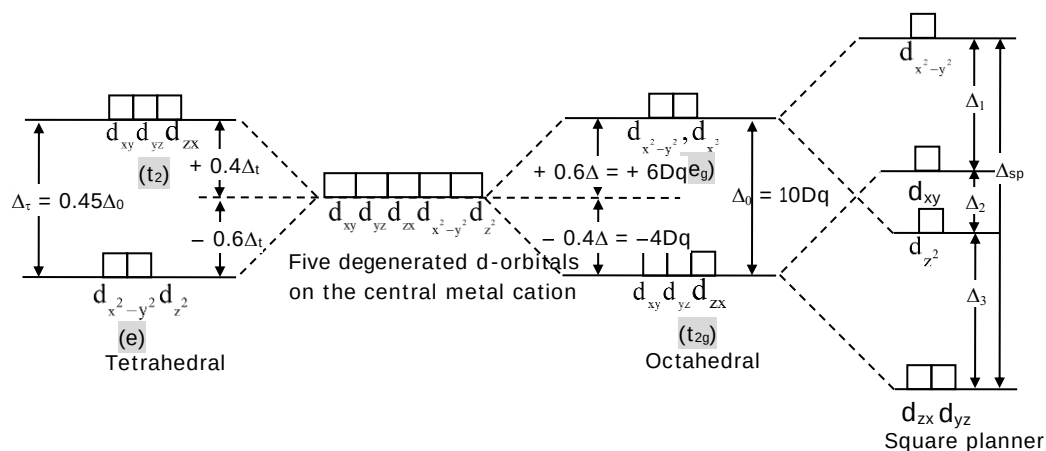
$$\text{Formula: CFSE} = [-0.6 n_{e_g} + 0.4 n_{t_{2g}}] \Delta_t + xP.$$

where $n_{t_{2g}}$ & n_{e_g} are number of electron(s) in $n_{t_{2g}}$ & n_{e_g} orbitals respectively and Δ_t crystal field splitting energy for tetrahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

(c) Crystal field splitting in square planar co-ordination entities

The square planar arrangement of ligands may be considered to be one derived from the octahedral field by removing two trans-ligands located along the Z-axis. In the process, the e_g and t_{2g} sets of orbitals is lifted i.e., these orbitals will no longer be degenerate.

The four ligands in square planar arrangement around the central metal ion are shown in Fig. As the ligands approach through the x and y axis, they would have greatest influence on $d_{x^2-y^2}$ orbital, so the energy of this orbital, will be raised most. The d_{xy} orbital, lying in the same plane, but between the ligands will also have a greater energy though the effect will be less than that on the $d_{x^2-y^2}$ orbitals. On the other hand, due to absence of ligands along Z-axis, the d_{z^2} orbital becomes stable and has energy lower than that of d_{xy} orbital. Similarly d_{yz} and d_{xz} become more stable. The energy level diagram may be represented as shown in figure along with tetrahedral and octahedral fields.



The value of Δ_{sp} has been found larger than Δ_o because of the reason that d_{xz} and d_{yz} orbitals interact with only two ligands in the square planar complexes, while in octahedral complexes the interaction takes place only with four ligands. Δ_{sp} has been found equal to $1.3\Delta_o$. Thus,

$$\Delta_{sp} = (\Delta_1 + \Delta_2 + \Delta_3) > \Delta_o \text{ and } \Delta_{sp} = 1.3 \Delta_o.$$

(E) Stability Of Coordination Compounds

The stability of a coordination compound $[ML_n]$ is measured in terms of the stability constant (equilibrium constant) given by the expression,

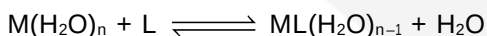
$$\beta_n = [ML_n] / [M(H_2O)_n][L]^n$$

for the overall reaction:

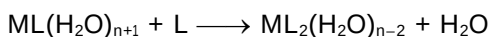


By convention, the water displaced is ignored, as its concentration remains essentially constant.

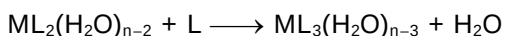
The above overall reaction takes place in steps, with a stability (formation) constant, $K_1, K_2, K_3, \dots, K_n$ for each step as represented below:



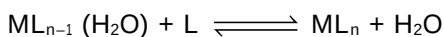
$$K_1 = [ML(H_2O)_{n-1}] / \{[M(H_2O)_n][L]\}$$



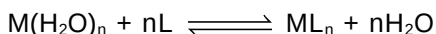
$$K_2 = [ML_2(H_2O)_{n-2}] / \{[ML(H_2O)_{n-1}][L]\}$$



$$K_3 = [ML_3(H_2O)_{n-3}] / \{[ML_2(H_2O)_{n-2}][L]\}$$



$$K_n = [ML_n] / \{[ML_{n-1}(H_2O)][L]\}$$



$$\beta_n = K_1 \times K_2 \times K_3 \times \dots \times K_n$$

β_n , the stability constant, is related to thermodynamic stability when the system has reached equilibrium. Most of the measurements have been made from aqueous solutions, which implies



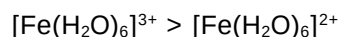
that the complex is formed by the ligand displacing, water from the aqua complex of the metal ion. Ignoring the charge and taking L as an unidentate ligand, the stepwise formation of the complex is represented as shown above.

$K_1, K_2, K_3 \dots K_n$ representing the stepwise stability (or formation) constants.

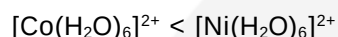
The above is thermodynamic stability criteria, there can be another kind of stability called kinetic stability, which measures the rate of ligand replacement.

(F) Factors Influencing The Magnitude Of C.F.S.E.

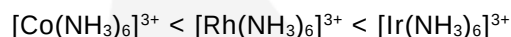
(a) Different charges on the cation of the same metal: The cation with a higher oxidation state has a larger value of CFSE than that with lower oxidation state e.g.,



(b) Same charges on the cation but the number of d-electrons are different: The metal cation the magnitude of CFSE with the increase of the number of d-electrons, e.g.,



(c) Quantum number (n) of the d-orbitals of the central metal ion: As 'n' increase CFSE increases.

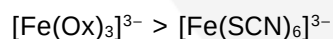


(d) Types of Hybridisation: $\Delta_t = \frac{4}{9} \Delta_0$

The order of CFSE:-

$$\Delta_{\text{sq}} > \Delta_0 > \Delta_t$$

(e) Presence of chelating ligand increases CFSE



Concept Builders



Q.21 Among the following ions, which one has the highest paramagnetism?

- (A) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (C) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (D) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$

Q.22 Which of the following complex involves d^2sp^3 hybridisation?

- (A) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{CoF}_6]^{3-}$ (C) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (D) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

Q.23 In Ziese's salt C = C bond length is:

$$\text{Note : } \begin{cases} \text{C - C both length in ethane is } 1.54 \text{ \AA} \\ \text{C = C both length in ethene is } 1.34 \text{ \AA} \\ \text{C} \equiv \text{C both length in ethyne is } 1.20 \text{ \AA} \end{cases}$$

- (A) 1.37 Å (B) 1.19 Å (C) 1.87 Å (D) 1.34 Å

Q.24 The ligand that shows synergic effect is:

- (A) H_2O (B) NH_3 (C) CO (D) None

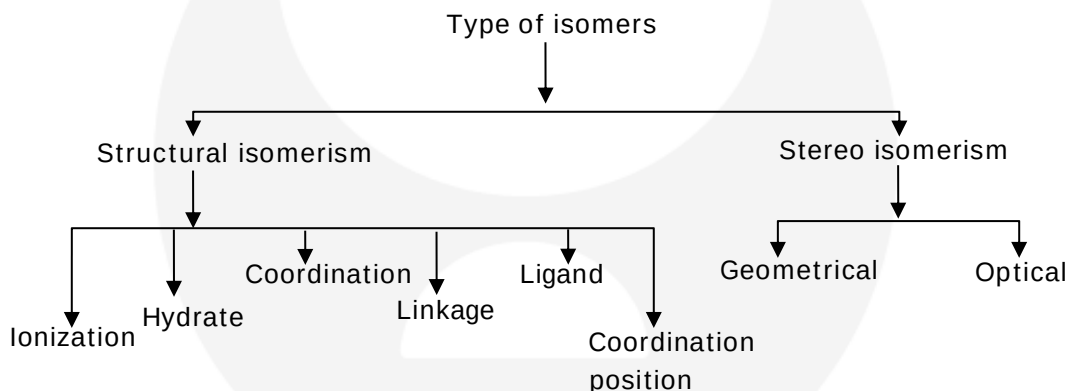
Q.25 All the metal ions contains configurations. Which of the following complex will be paramagnetic?

- (A) $[\text{FeCl}(\text{CN})_4(\text{O}_2)]^{4-}$ (B) $\text{K}_4[\text{Fe}(\text{CN})_6]$
(C) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (D) $[\text{Fe}(\text{CN})_5(\text{O}_2)]^{-5}$

Isomerism In Complexes

- (a) Compounds which have the same molecular formula, but differ in their properties due to the difference in structure are called as Isomers.
- (b) Isomerism is commonly considered, to be the characteristic of only organic compounds, it is also found although less frequently among inorganic substances.

Classification Of Isomerism



(A) Structural isomerism

It arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere.

(a) Ionisation isomerism

The type of isomerism which is due to the exchange of groups or ions between the coordination sphere and the ionisation sphere.

Example:

- (i) $\text{Co}(\text{NH}_3)_4 \text{Br}_2 \text{SO}_4$ can be represented as
 $[\text{Co}(\text{NH}_3)_4 \text{Br}_2] \text{SO}_4$ (red violet) and $[\text{Co}(\text{NH}_3)_4 \text{SO}_4] \text{Br}_2$ (red)
These complexes give sulphate ion and bromide ion respectively
- (ii) $[\text{Pt}(\text{NH}_3)_4 \text{Cl}_2] \text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4 \text{Br}_2] \text{Cl}_2$
- (iii) $[\text{Co}(\text{NH}_3)_4 (\text{NO}_3)_2] \text{SO}_4$ and $[\text{Co}(\text{NH}_3)_4 \text{SO}_4] (\text{NO}_3)_2$



(b) Hydrate isomerism

This type of isomerism is due to presence of different number of water molecules inside a coordination sphere.

Example:

$\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3$ has four possible structures

- (i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet (ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ green
(iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green (iv) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ dark green.

These complexes differ from one another with respect to the number of water molecules acting as ligands. Other hydrate isomers are

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$

(c) Linkage isomerism

- (i) This type of isomerism arises due to presence of ambidentate ligands like NO_2^- , CN^- and SCN^-
(ii) These ligands have two donor atoms but at a time only one atom is directly linked to the central metal atom of the complex.
(iii) Such type of isomers are distinguished by infra red (I.R.) spectroscopy.

Example:

$[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$

In NO_2^- ligand, The coordinating sites are nitrogen i.e., NO_2^- or through oxygen i.e. ONO^-
The nitro isomer is yellow and is stable to acids whereas nitrito isomer is red and is decomposed by acids.

(d) Coordination isomerism

- (i) This type of isomerism is exhibited when the complex has two complex ions in it 'cationic and anionic'.
(ii) This type of isomerism is caused by the interchange of ligands between the two complex ions of the same complex.

Example:

$[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$ and $[\text{Co}(\text{NH}_3)_5(\text{CN})] [\text{Cr}(\text{CN})_5(\text{NH}_3)]$

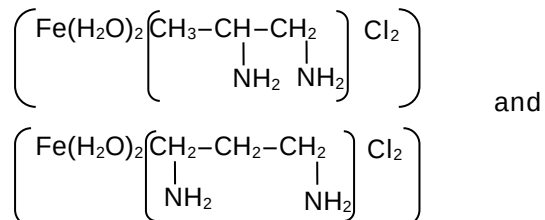
(e) Ligand isomerism

- (i) Ligands like $\text{C}_3\text{H}_6(\text{NH}_2)_2$ have two different structures i.e. 1, 3-diamino propane and 1, 2-diaminopropane(propylene diamine).
(ii) Those complexes which have same molecular formula, but differ with respect to their ligands are called as Ligand isomers.



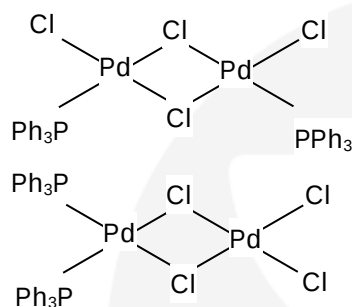
Example:

$[\text{Fe}(\text{H}_2\text{O})_2 \text{C}_3\text{H}_6(\text{NH}_2)\text{Cl}_2]$ has two different structures



(f) Co-ordination position isomerisation

It is shown by polynuclear complexes, due interchange of ligands between the different metal nuclei.



(g) Polymerization Isomerism

This is not true isomerism because it occurs between compounds having the same empirical formula, but different molecular weights.

Example:

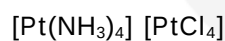
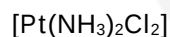


Illustration:

Select correct code about complex $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5][\text{ZnCl}_4]$

(I) IUPAC name of the compound is Pentaamminenitrito-N-chromium(III) tetrachloridozincate (II)

(II) It shows geometrical isomerism

(III) It shows linkage isomerism

(IV) It shows co-ordination isomerism

(A) III, IV

(B) I, III & IV

(C) II, III & IV

(D) I, II, III & IV

Ans. (B)

Solution:

(I) $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5][\text{ZnCl}_4]$

IUPAC name of this compound is Pentaamminenitrito-N-chromium(III) tetrachloridozincate(II)



(III) It shows linkage isomerism due to presence of ambidentate ligands



(IV) Because both complex ions have different ligands

(B) Stereo isomerism

They have same molecular formula, same constitution, they differ only with respect to the spatial orientation of ligands in space around the metal ion. The two stereo isomers which are possible Geometrical and Optical.

(a) Geometrical isomerism

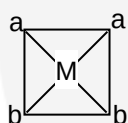
(i) The ligands occupy different positions around the central metal ion.

(ii) When two identical ligands are co-ordinated to the metal ion from same side then it is cis isomer. (Latin, cis means same).

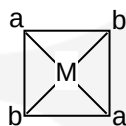
(iii) If the two identical ligands are co-ordinated to the metal ion from opposite side then it is trans isomer (in Latin, trans means across).

Geometrical isomers with co-ordination number = 4 (Square planar complexes)

(i) Complexes with general formula, Ma_2b_2 (where both a and b are monodentate) can have cis-and trans isomers.

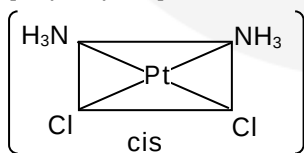
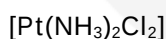


Cis-isomer

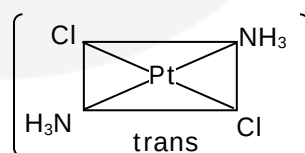


trans-isomer

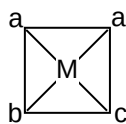
Example:



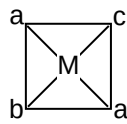
(cis-platin) anti cancer



(ii) Complexes with general formula Ma_2bc can have cis and trans-isomers.



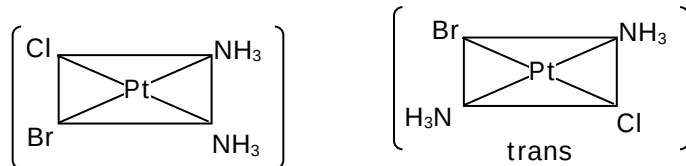
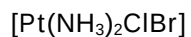
cis



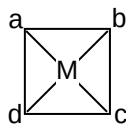
trans



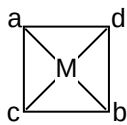
Example:



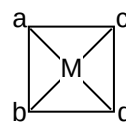
(iii) Complexes with general formula Mabcd can have cis and trans-isomers.



(i)

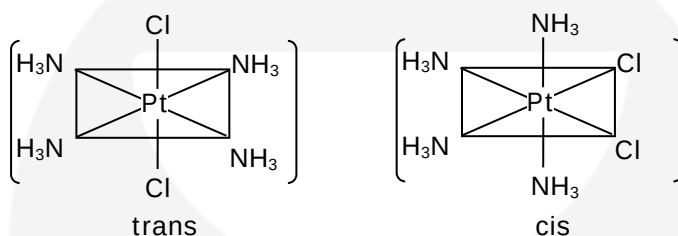
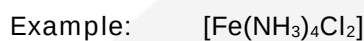


(ii)

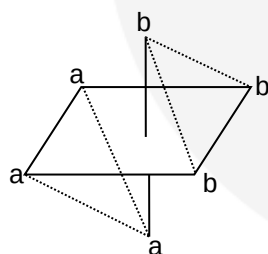


(iii)

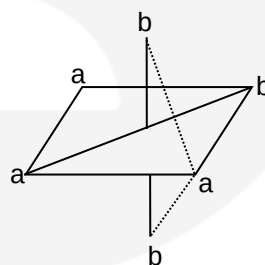
Geometrical isomers with co-ordination number = 6



(vi) Facial and Meridional isomerism (Ma_3b_3)



Facial (fac)



Meridional (Mer)

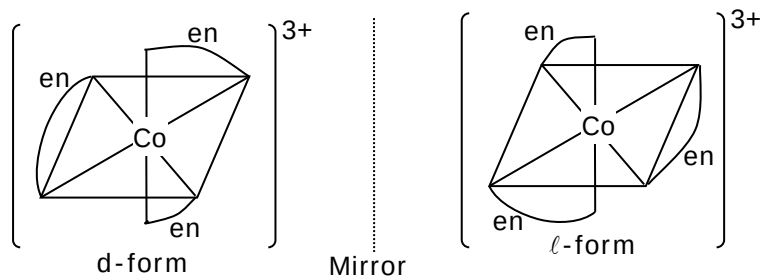
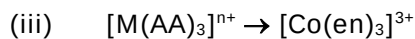
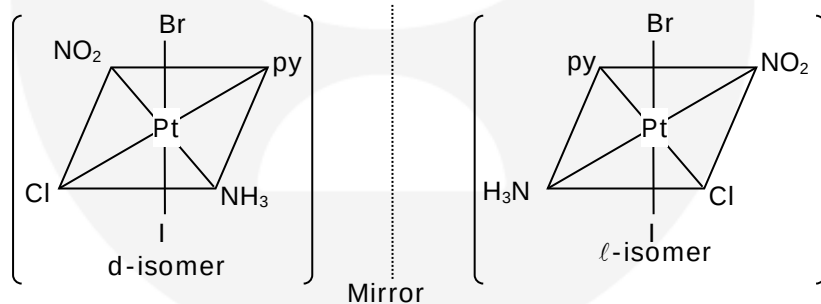
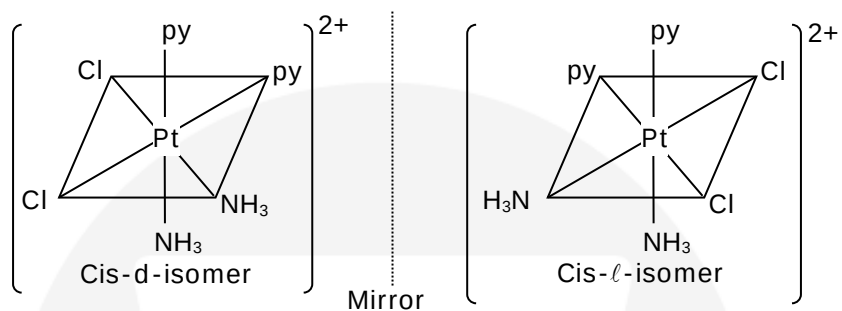
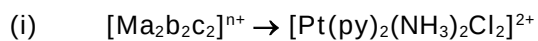
(b) Optical isomers

- (i) Optically active complexes are those which are nonsuperimposable over the mirror image structure.
- (ii) An optically active complex is one which is asymmetric in nature i.e., not divisible into two identical halves.
- (iii) The complex which rotates plane polarised light to left hand side is laevo rotatory i.e. ' ℓ ' or ' $-$ ' and if the complex rotates the plane polarised light to right hand side then it is dextro rotatory ' d ' or ' $+$ '.



- (iv) Thus complexes which have same physical and chemical properties but differ in their action towards plane polarised light are called as optical isomers.
- (v) The 'd' and 'l' isomers of a compound are called as Enantiomers or Enantiomorphs.
- (vi) Optical isomerism is expected in tetrahedral complexes of the type $Mabcd$.

Optical isomers with Co-ordination number = 6





Number of Possible Isomers For Specific Complexes
(For coordination number = 6)

Formula	Geometrical Isomers	Cis	Trans	Pairs of Enantiomers	Number of stereoisomers
Ma ₆	–	–	–	–	1
Ma ₅ b	–	–	–	–	1
Ma ₄ b ₂	2	1	1	–	2
Ma ₃ b ₃	2	Fac	Mer	–	2
Ma ₄ bc	2	1	1	–	2
Ma ₃ b ₂ c	3	1	2	–	3
Ma ₃ bcd	4	1	3	1	5
Ma ₂ b ₂ c ₂	5	1	4	1	6
Ma ₂ b ₂ cd	6	2	4	2	8
Ma ₂ bcde	9	6	3	6	15
Mabcdef	15	–	–	15	30
M(AA) ₃	–	–	–	1	2
M(AA) ₂ b ₂	2	1	1	1	3
M(AA) ₂ bc	2	1	1	1	3
M(AA)b ₄	–	–	–	–	1
M(AA)b ₃ c	2	1	1	–	2
M(AA)b ₂ c ₂	3	1	2	1	4
M(AA)b ₂ cd	4	2	2	2	6
M(AA)bcde	6	–	–	6	12
M(AB) ₃	2	1	1	2	4
M(AB) ₂ c ₂	5	1	4	3	8
M(AB) ₂ cd	6	2	4	5	11
M(AB)c ₄	–	–	–	–	1
M(AB)c ₃ d	3	1	2	1	4
M(AB)c ₂ d ₂	4	2	2	2	6
M(AB)c ₂ de	7	4	3	5	12
M(AB)cdef	12	–	–	12	24
M(AA)(BC)de	5	–	–	5	10
M(AB)(CD)ef	10	–	–	10	20



Number Of Possible Isomers For Specific Square Planar Complexes
(For coordination number = 4)

Formula	Geometrical Isomers	Cis	Trans	Number of stereoisomers
Ma_4	–	–	–	1
Ma_3b	–	–	–	1
Ma_2b_2	2	1	1	2
Ma_2bc	2	1	1	2
Mabcd	3	–	–	3
$\text{M}(\text{AA})_2$	–	–	–	1
$\text{M}(\text{AA})\text{b}_2$	–	–	–	1
$\text{M}(\text{AA})\text{bc}$	–	–	–	1
$\text{M}(\text{AB})_2$	2	1	1	2
$\text{M}(\text{AB})\text{c}_2$	–	–	–	1
$\text{M}(\text{AB})\text{cd}$	2	–	–	2

Note: Uppercase letters represent chelating ligands and lowercase letters represent monodentate ligands.

(AA) → Symmetrical bidentate ligand

(AB)/(BC)/(CD) → unsymmetrical bidentate ligand

Concept Builders



Q.26 1 Which of the following complexes shows ionization isomerism?

(A) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$

(B) $[\text{Cr}(\text{en})_2]\text{Cl}_2$

(C) $[\text{Cr}(\text{en})_3]\text{Cl}_3$

(D) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

Q.27 The complex $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ and $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ are called

(A) ionisation isomer

(B) linkage isomers

(C) coordination isomer

(D) geometrical isomer

Q.28 The two compounds pentaamminesulphatocobalt (III) bromide and entaamminesulphatocobalt (III) chloride represent:

(A) Linkage isomerism

(B) Ionization isomerism

(C) Coordination isomerism

(D) No isomerism

Q.29 Geometrical isomerism is found in coordination compounds having coordination number:

- (A) 2 (B) 3
(C) 4 (tetrahedral) (D) 6

Q.30 Geometrical isomerism can be shown by:

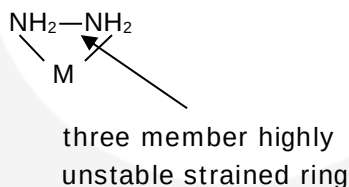
- (A) $[\text{Ag}(\text{NH}_3)(\text{CN})]$ (B) $\text{Na}_2[\text{Cd}(\text{NO}_2)_4]$
(C) $[\text{PtCl}_4\text{I}_2]$ (D) $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{Au}(\text{CN})_4]$

Solved Examples

1. NH_2NH_2 , although possess two electron pair for donation but not acts as chelating agent. Why?

Solution:

The coordination by NH_2NH_2 , leads to a three member highly unstable strained ring and thus it does not acts as chelating agent.



2. Square planar complexes with co-ordination number four exhibit geometrical isomerism whereas tetrahedral complexes do not. Why ?

Solution:

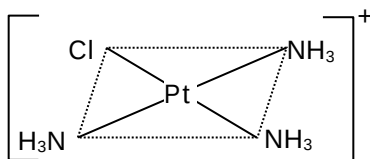
In tetrahedral complexes, the relative position of atoms with respect to each other is same thus these do not show geometrical isomerism. Square planar complexes show cis, trans isomerism.

3. Platinum (II) forms square planar complexes and platinum (IV) gives octahedral complexes. How many geometrical isomers are possible for each of the following complexes ? Describe their structures -

- (A) $[\text{Pt}(\text{NH}_3)\text{Cl}]^+$ (B) $[\text{Pt}(\text{NH}_3)\text{Cl}_5]^-$ (C) $[\text{Pt}(\text{NH}_3)_2\text{CNO}_2]$ (D) $[\text{Pt}(\text{NH}_3)_4\text{ClBr}]^{2+}$

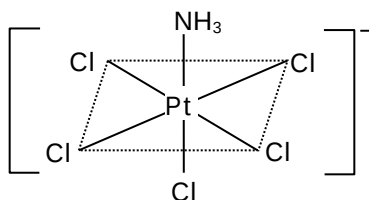
Solution:

(A) Square planar complex of the type Ma_3b shows no isomerism.

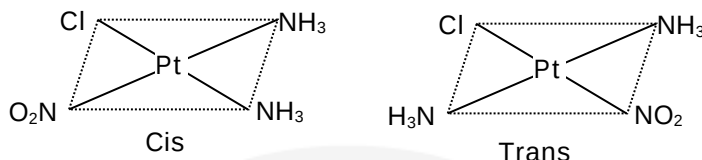




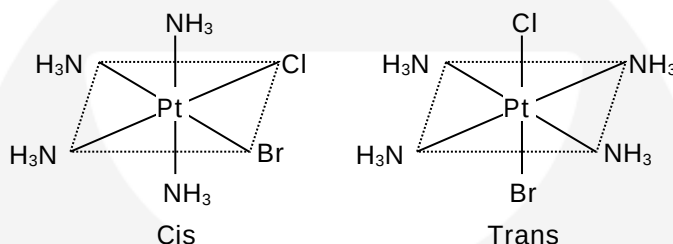
(B) an octahedral complex of the type Ma_b5 shows no isomerism.



(C) A square planar complex of the type Ma_2b_2 exists as cis and trans isomers



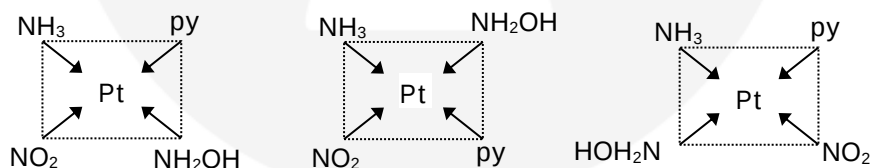
(D) An octahedral complex of the type Ma_4bc shows cis and trans isomerism.



4. How many geometrical isomers with formula $[MABCD]$ e.g., $[Pt(NH_3)py(NH_2OH)NO_2]^+$ are possible?

Solution:

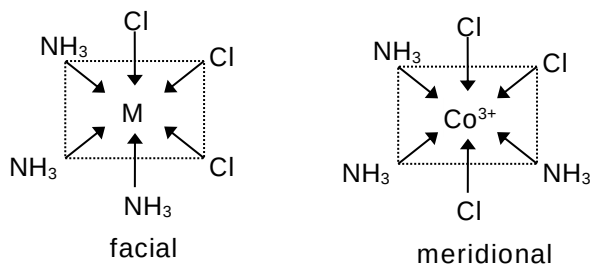
Three isomers



5. What are facial and meridional isomers? Explain with examples.

Solution:

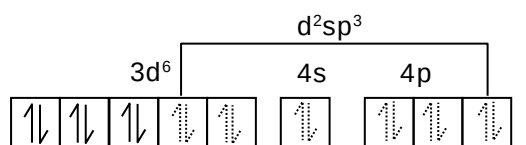
For the complexes of the type $[Ma_3b_3]$ e.g., $[Co(NH_3)_3Cl_3]$ and $[Rh(py)_3Cl_3]$; the isomer is called facial (fac) when all similar ligands occupy same face of an octahedron as shown below. When all similar ligands are not on the same face then isomer is called meridional (mer).



6. $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic while $[\text{FeF}_6]^{4-}$ is strongly paramagnetic. Why?

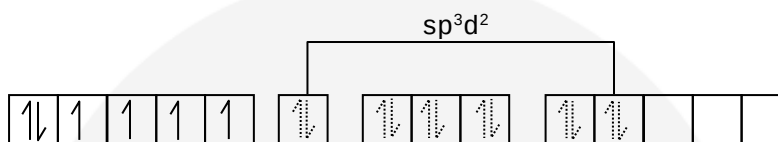
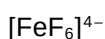
Solution:

CN^- is strong field ligand and so it pairs up the electrons leading to the formation of low spin complex which is diamagnetic.



Octahedral, diamagnetic complex

F^- is weak field ligand so it forms high spin paramagnetic complex.



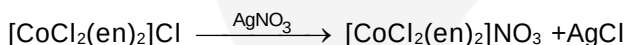
Octahedral, paramagnetic complex

7. If excess of AgNO_3 solution is added to 100 mL of a 0.024 M solution of dichlorobis (ethylene diamine) cobalt(III) chloride, how many mol of AgCl will be precipitated:

(A) 0.0012 (B) 0.0016 (C) 0.0024 (D) 0.0048

Ans. (C)

Solution:



Mole of $[\text{CoCl}_2(\text{en})_2]\text{Cl}$ = molarity $\times$ v(Lt.)

0.0024 mole of $[\text{CoCl}_2(\text{en})_2]\text{Cl}$ gives

$$\frac{100}{1000} \times 0.024 = 0.0024 \text{ mole of } \text{Cl}^-$$

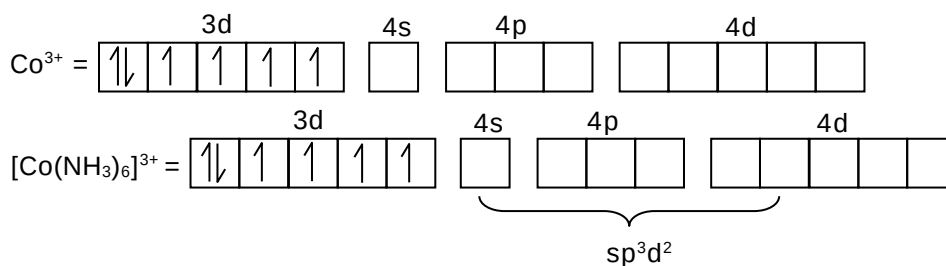
Which precipitate 0.0024 mole of AgNO_3 because 1 mole Ag^+ required 1 mole Cl^- to precipitate

8. Find out the hybridization, geometry and magnetic moment of the complexes:

(i) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (ii) $[\text{Cr}(\text{CN})_6]^{3-}$

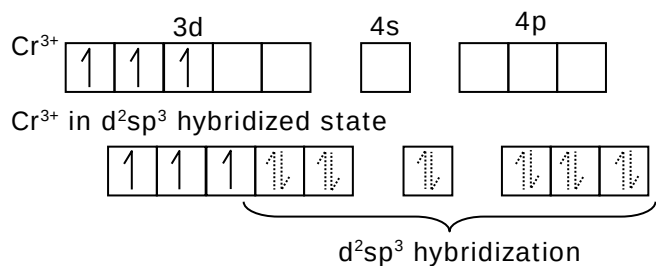
Solution:

- (i) The oxidation state of cobalt in the complex is +3. The electronic configuration of Co^{3+} ions is





- (ii) The oxidation state of chromium in the complex is +3. The electronic configuration of Cr^{3+} ion is



Octahedral, magnetic moment = $\sqrt{3 \times (3 + 2)} = \sqrt{15} = 3.87 \text{ B.M.}$

9. The magnetic moment of $[\text{MnCl}_4]^{2-}$ is 5.92 B.M. On the basis of its magnetic moment, write configuration of Mn^{2+} in this complex.

Solution:

For an atom/ion

Magnetic moment (μ) = $\sqrt{n(n+2)}$

(n = No. of unpaired electrons)

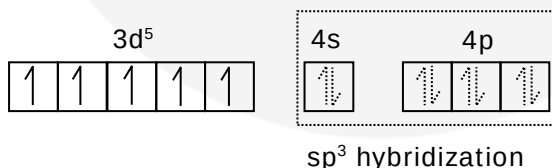
Given that $\mu = 5.92 \text{ B.M.}$

$\therefore 5.92 = \sqrt{n(n+2)}$

or $n = 5$

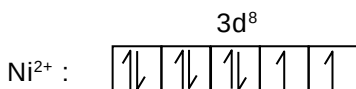
Thus in this complex Mn contains five unpaired electrons and so its possible configuration may be Mn^{2+} in $[\text{MnCl}_4]^{2-} = [\text{Ar}]3d^5 4s^0$

so hybridisation of Mn^{2+} in the given complex must be sp^3 .



10. All the octahedral complexes of Ni^{2+} are outer orbital complexes, why ?

Solution:



Thus only one vacant 3d-orbital is available after pairing up of electrons due to strong field ligand.

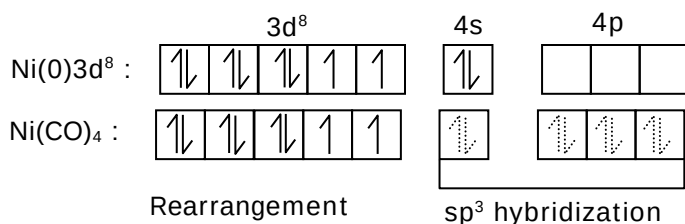
Therefore, d^2sp^3 hybridisation is not possible. Only sp^3d^2 is possible which represents outer complex.



11. $\text{Ni}(\text{CO})_4$ possesses tetrahedral geometry while $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar, why?

Solution:

In the formation of $\text{Ni}(\text{CO})_4$, nickel undergoes sp^3 hybridization, hence it is tetrahedral in shape.



12. Consider the following complexes-



Their electrical conductance's in aq. solutions are -

(A) 256, 0, 97, 404

(B) 404, 0, 97, 256

(C) 256, 97, 0, 404

(D) 404, 97, 256, 0

Solution:

(A) Coordination number of Pt is 6 hence

I $\text{K}_2[\text{PtCl}_6]$ – Three ions

II $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ – Zero ions

III $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$ – Two ions

IV $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ – Four ions

Conductivity $\propto$ no. of ions

13. Determine the oxidation state of metal in the complex ion, $[\text{PtCl}_6]^{2-}$

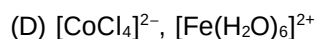
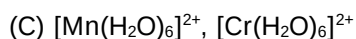
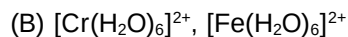
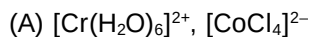
Solution:

Charge on the complex ion = Oxidation state of metal + charge on ligands

$$-2 = x + 6 \times (-1) \text{ or } x = +4$$

The oxidation of Pt in the complex ion is +4.

14. The pair in which both species have same magnetic moment (spin only value) is-

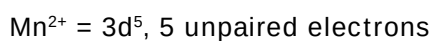


Ans. (B)

Solution:

$$\text{Same magnetic moment} = \text{same number of unpaired electrons} = \sqrt{n(n+2)}$$

where n = number of unpaired electrons

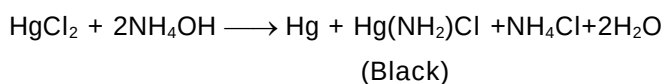




15. The colour of Hg_2Cl_2 changes from white to black when treated with NH_3 . Why ?

Solution:

Hg_2Cl_2 reacts with NH_4OH , to give black complex.



16. Write the formula of the following complexes.

(i) Pentamminechlorocobalt (III) ion

(ii) Lithium tetrahydridoaluminate (III)

Solution:

(i) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$;

(ii) $\text{Li}[\text{AlH}_4]$

17. Identify the complexes which are expected to be coloured and explain.

(A) $\text{Ti}(\text{NO}_3)_4$

(B) $[\text{Cu}(\text{NCCH}_3)_4]^+\text{BF}_4^-$

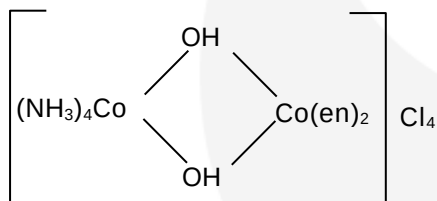
(C) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$

(D) $\text{K}_3[\text{VF}_6]$

Solution:

(C) and (D) Are coloured because Cr^{3+} in $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and V^{3+} in $[\text{VF}_6]^{3-}$ have $3d^3$ and $3d^2$ configuration respectively and thus show d-d transition.

18. Write the IUPAC name of the given compound.



Solution:

Tetraamine cobalt (III) di- μ -hydroxo bis ethylenediamine cobalt (III) chloride.

19. Explain the following with appropriate reasons:

$[\text{Fe}(\text{CN})_6]^{3-}$ ion has magnetic moment 1.73 BM, while $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has a magnetic moment 5.92 B.M.

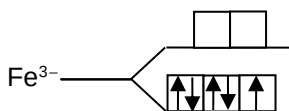
Solution:

$$1.73 = \sqrt{n(n+2)}$$

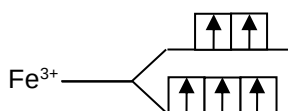
$$n = 1$$

$$5.92 = \sqrt{n(n+2)}$$

$$n = 5$$



Due to strong field ligands (CN^-)



(Due to weak field ligands (H_2O))



ANSWER KEY FOR CONCEPT BUILDER

1.	(D)	2.	(A)	3.	(B)	4.	(D)	5.	(B)	6.	(D)	7.	(C)
8.	(A)	9.	(C)	10.	(B)	11.	(B)	12.	(C)	13.	(C)	14.	(D)
15.	(D)	16.	(D)	17.	(A)	18.	(C)	19.	(D)	20.	(C)	21.	(B)
22.	(A)	23.	(A)	24.	(C)	25.	(A)	26.	(D)	27.	(B)	28.	(D)
29.	(D)	30.	(C)										





Single Correct Type Question

General Problems on Coordination Chemistry

- The oxidation state of Mo in its oxido-complex species $[\text{Mo}_2\text{O}_4(\text{C}_2\text{H}_4)_2(\text{H}_2\text{O})_2]^{2-}$ is
(A) +2 (B) +3 (C) +4 (D) +5
 - Consider the complex ion $[\text{Cr}(\text{NH}_3)_2(\text{C}_2\text{O}_4)_2]^-$. What is the?
(i) oxidation number of the metal atom.
(ii) coordination number of the metal atom.
(iii) charge on the complex if all ligands were chloride ions
(Consider no change in co-ordination number)
- | | (i) | (ii) | (iii) |
|-----|-----|------|-------|
| (A) | -1 | 6 | -3 |
| (B) | +3 | 6 | -3 |
| (C) | +3 | 6 | -1 |
| (D) | -1 | 5 | -3 |
- Which of the following is copper(I) compound?
(A) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (B) $[\text{Cu}(\text{CN})_4]^{3-}$ (C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (D) All of these
 - Some salts although containing two different metallic elements give test for one of them in solution. Such salts are
(A) complex salt (B) double salt (C) normal salt (D) none of these
 - Aqueous solution of FeSO_4 gives tests for both Fe^{2+} and SO_4^{2-} but after addition of excess of KCN, solution ceases to give test for Fe^{2+} . This is due to the formation of
(A) the double salt $\text{FeSO}_4 \cdot 2\text{KCN} \cdot 6\text{H}_2\text{O}$ (B) $\text{Fe}(\text{CN})_3$
(C) the complex ion $[\text{Fe}(\text{CN})_6]^{4-}$ (D) the complex ion $[\text{Fe}(\text{CN})_6]^{3-}$

Ligands

- Ethylene diamine is an example of a ligand:
(A) monodentate (B) bidentate (C) tridentate (D) hexadentate
- How many EDTA^{4-} molecules are required to make an octahedral complex with a Ca^{2+} ion?
(A) Six (B) Three (C) One (D) Two
- The donor sites of $(\text{EDTA})^{4-}$ are?
(A) O atoms only (B) N atoms only
(C) Two N atoms and four O atoms (D) Three N atoms and three O atoms



9. Diethylene triamine is:
(A) Chelating agent (B) Polydentate ligand
(C) Tridentate ligand (D) All of these
10. Which of the following has five donor (coordinating) sites?
(A) Triethylene tetramine (B) Ethylenediamine tetracetate ion
(C) Ethylenediamine triacetate ion (D) Diethylene triamine

Synergic Bonding

11. Which of the following ligands is called π -acceptors?
CO CN⁻ NO⁺
(I) (II) (III)
(A) I, II, III only correct. (B) I, II only correct
(C) II, III only correct (D) III only correct
12. In Fe(CO)₅, the Fe–C bond possesses:
(A) π character only (B) σ character only
(C) ionic character only (D) both π and σ characters
13. π -bonding is not involved in:
(A) ferrocene (B) dibenzene chromium
(C) Zeise's salt (D) Grignard reagent
14. Which of the following is not considered as an organometallic compound?
(A) Ferrocene (B) Cis-platin (C) Ziese's salt (D) Grignard reagent
15. Formula of ferrocene is:
(A) [Fe(CN)₆]⁴⁻ (B) [Fe(CN)₆]³⁺ (C) [Fe(CO)₅] (D) [Fe(C₅H₅)₂]

Effective Atomic Number (EAN) & **Werner's Theory**

16. Which of the following obey Sidgwick effective atomic number rule except
(A) [Cr(CO)₆] (B) [Co(NH₃)₆]³⁺ (C) [Ni(NH₃)₆]²⁺ (D) [PtCl₆]²⁻
17. EAN of a metal carbonyl M(CO)_x is 36. If atomic number of metal M is 26, what is the value of x?
(A) 4 (B) 8 (C) 5 (D) 6
18. Effective atomic number of Co(CO)₄ is 35, hence it is less stable. It attains stability by
(A) Oxidation of Co (B) Reduction of Co
(C) Dimerization (D) Both (B) & (C)



19. A compound is made by mixing cobalt (III) nitrite and potassium nitrite solutions in the ratio of 1 : 3. The aqueous solution of the compound showed 4 particles per molecule whereas molar conductivity reveals the presence of six electrical charges. The formula of the compound is:
(A) $\text{Co}(\text{NO}_2)_3 \cdot 2\text{KNO}_2$ (B) $\text{Co}(\text{NO}_2)_3 \cdot 3\text{KNO}_2$ (C) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ (D) $\text{K}[\text{Co}(\text{NO}_2)_4]$
20. Consider the following statements:
According to the Werner's theory.
(a) Ligands are connected to the metal ions by covalent bonds.
(b) Secondary valencies have directional properties
(c) Secondary valencies are non-ionisable
Of these statements:
(A) a, b and c are correct (B) b and c are correct
(C) a and c are correct (D) a and b are correct
21. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is:
(A) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_4$ (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ (C) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ (D) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
22. The molar ionic conductances of the octahedral complexes.
(I) $\text{PtCl}_4 \cdot 5\text{NH}_3$ (II) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (III) $\text{PtCl}_4 \cdot 3\text{NH}_3$ (IV) $\text{PtCl}_4 \cdot 2\text{NH}_3$
(A) $\text{I} < \text{II} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{III} < \text{II} < \text{I}$ (C) $\text{III} < \text{IV} < \text{II} < \text{I}$ (D) $\text{IV} < \text{III} < \text{I} < \text{II}$
23. On adding AgNO_3 solution to a solution of $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$, the percentage of total chloride ion precipitated is:
(A) 100 (B) 75 (C) 50 (D) 25

IUPAC

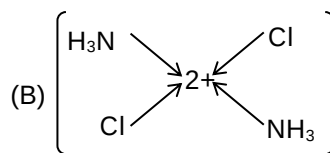
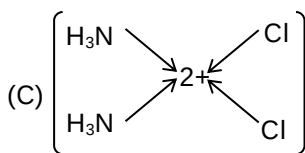
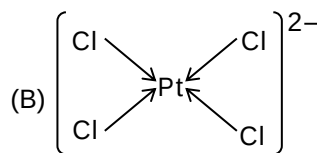
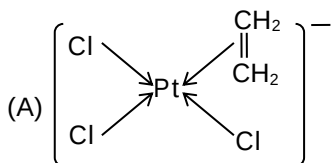
24. A complex cation is formed by Pt (in some oxidation state) with ligands (in proper number so that coordination number of Pt becomes six). Which of the following can be its correct IUPAC name?
(A) Diammineethylenediaminedithiocyanato-S-platinum (II) ion
(B) Diammineethylenediaminedithiocyanato-S-platinate (IV) ion
(C) Diammineethylenediaminedithiocyanato-S-platinum (IV) ion
(D) Diamminebis (ethylenediamine) dithiocyanate-S- platinum (IV) ion
25. The correct IUPAC name for the compound $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{ONO})]\text{Cl}$ is:
(A) Tetraamminechloridonitrito-N-cobalt(III) chloride
(B) Chloridonitrito-O-tetraamminecobalt(II) chloride
(C) Dichloridonitrito-O-tetraamminecobalt(III)
(D) Tetraamminechloridonitrito-O-cobalt(III) chloride



26. Trioxalatoaluminate(III) and tetrafluoro-borate(III) ions are respectively
(A) $[\text{Al}(\text{C}_2\text{O}_4)_3]$, $[\text{BF}_4]^{3-}$ (B) $[\text{Al}(\text{C}_2\text{O}_4)_3]^{3+}$, $[\text{BF}_4]^{3+}$
(C) $[\text{Al}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{BF}_4]^-$ (D) $[\text{Al}(\text{C}_2\text{O}_4)_3]^{2-}$, $[\text{BF}_4]^{2-}$
27. The formula of the complex hydridotrimethoxidoborate(III) ion is:
(A) $[\text{BH}(\text{OCH}_3)_3]^{2-}$ (B) $[\text{BH}_2(\text{OCH}_3)_3]^{2-}$ (C) $[\text{BH}(\text{OCH}_3)_3]^-$ (D) $[\text{BH}(\text{OCH}_3)_3]^+$
28. The IUPAC name of the Wilkinson's catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is
(A) Chloridotris(triphenylphosphine)rhodium(I)
(B) Chloridotris(triphenylphosphine)rhodium(IV)
(C) Chloridotris(triphenylphosphine)rhodium(0)
(D) Chloridotris(triphenylphosphine)rhodium(VI)
29. The formula for the compound tris (ethane-1, 2-diamine)cobalt(III) sulphate is
(A) $[\text{Co}(\text{en})_3]\text{SO}_4$ (B) $[\text{Co}(\text{SO})_4(\text{en})_3]$ (C) $[\text{Co}(\text{en})_3](\text{SO}_4)_2$ (D) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$

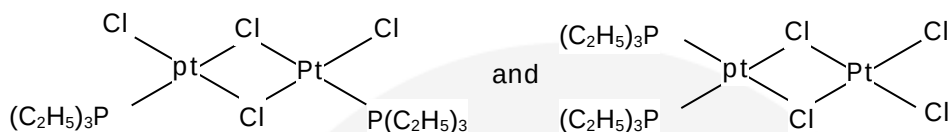
Isomerism

30. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ are related to each other as:
(A) geometrical isomers (B) linkage isomers
(C) coordination isomers (D) ionisation isomers
31. In which of the following pairs both the complexes show optical isomerism?
(A) $\text{cis}-[\text{Cr}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3-}$, $\text{trans}-[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (B) $[\text{Co}(\text{en})_3]\text{Cl}_3$, $\text{cis}-[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
(C) $[\text{PtCl}(\text{dien})]\text{Cl}$, $[\text{NiCl}_2\text{Br}_2]^{2-}$ (D) $[\text{Co}(\text{NO}_3)_3(\text{NH}_3)_3]$, $\text{cis}-[\text{Pt}(\text{en})_2\text{Cl}_2]$
32. Which of the following does not have optical isomer?
(A) $[\text{Co}(\text{en})_3]\text{Cl}_3$ (B) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (C) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (D) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]\text{Cl}$
33. Which of the following is considered to be an anticancer species?

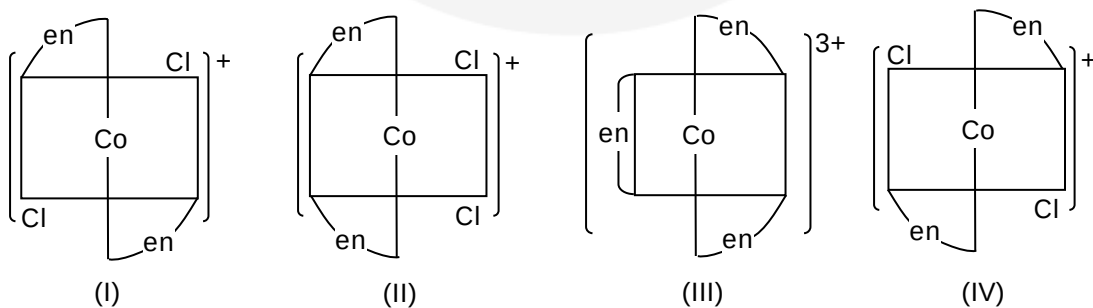




34. Which of the following can exhibit geometrical isomerism?
(A) $[\text{MnBr}_4]^{2-}$ (B) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$
(C) $[\text{PtCl}_2\{\text{P}(\text{C}_2\text{H}_5)_3\}_2]$ (D) $[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]^{2+}$
35. Type of isomerism exhibited by $[\text{Cr}(\text{NCS})(\text{NH}_3)_5]$ $[\text{ZnCl}_4]$
(A) Coordination isomerism (B) Linkage isomerism
(C) Ionization isomerism (D) Both coordination and linkage isomerism
36. The complexes given below show:



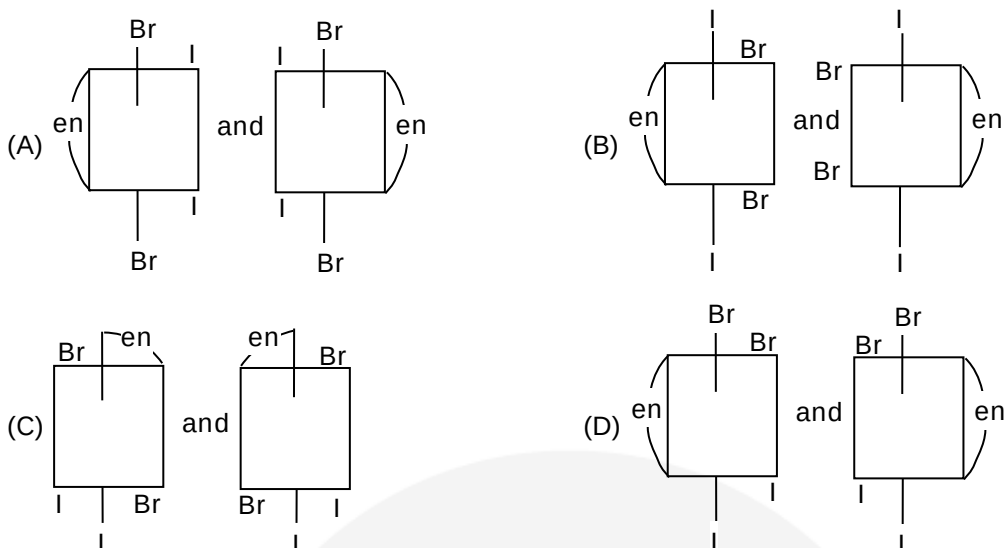
- (A) Optical isomerism (B) Co-ordination isomerism
(C) Geometrical isomerism (D) Co-ordination position isomerism
37. The total number of possible isomers of the compound $[\text{Cu}^{\text{II}}(\text{NH}_3)_4] [\text{Pt}^{\text{II}}\text{Cl}_4]$ are:
(A) 3 (B) 5 (C) 4 (D) 6
38. Which complex is likely to show optical activity:
(A) Trans- $[\text{CoCl}_2(\text{NH}_3)_4]^+$ (B) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
(C) Cis- $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (D) Trans- $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$
39. Which of the following ions are optically active?



- (A) I only (B) II only (C) II and III (D) IV only



40. The complex ion has two optical isomers. Their correct configurations are:



VB T

41. The structure of iron pentacarbonyl is:
(A) Square planar (B) Trigonal bipyramid (C) Triangular (D) None of these
42. The complex $[\text{Pt}(\text{NH}_3)_4]^{2+}$ has structure:
(A) square planar (B) tetrahedral (C) pyramidal (D) pentagonal
43. $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{NH}_3)_4]^{2+}$ do not differ in
(A) magnetic moment (B) oxidation number of Ni
(C) geometry (D) EAN
44. For the correct assignment of electronic configuration of a complex, the valence bond theory often requires the measurement of
(A) molar conductance (B) optical activity
(C) magnetic moment (D) dipole moment
45. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal ion could be
(A) Co^{2+} (B) Mn^{2+} (C) Fe^{2+} (D) Fe^{3+}

CFT

46. The magnitude of crystal field stabilisation energy in octahedral field depends on
I: the nature of the ligand
II: the charge on the metal ion
III: whether the metal is in the first, second or third row of the transition elements.
(A) I, II, III are correct. (B) I, II are correct
(C) II, III are correct (D) III is only correct



47. Which of the following statements is correct?
(A) $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ both are paramagnetic complexes.
(B) $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ both are high spin complexes.
(C) $[\text{CoF}_6]^{3-}$ is octahedral while $[\text{Co}(\text{NH}_3)_6]^{3+}$ has a pentagonal pyramid shape.
(D) $[\text{CoF}_6]^{3-}$ is outerorbital complex while $[\text{Co}(\text{NH}_3)_6]^{3+}$ is inner orbital complex.
48. Which of the following is a high spin complex?
(A) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (B) $[\text{Fe}(\text{CN})_6]^{4-}$ (C) $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{FeF}_6]^{3-}$
49. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (atomic number of Cr = 24) has a magnetic moment of 3.83 B.M. The correct distribution of 3d electrons in the chromium present in the complex is:
(A) $3d_{xy}^1, 3d_{yz}^1, 3d_{zx}^1$ (B) $3d_{xy}^1, 3d_{yz}^1, 3d_{z^2}^1$
(C) $3d_{(x^2-y^2)}^1, 3d_{z^2}^1, 3d_{xz}^1$ (D) $3d_{xz}^1, 3d_{(x^2-y^2)}^1, 3d_{yz}^1$
50. The correct order of magnetic moments is:
(A) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$ (B) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
(C) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$ (D) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
51. Amongst $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ and NiCl_4^{2-} :
(A) $\text{Ni}(\text{CO})_4$ and NiCl_4^{2-} are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic.
(B) NiCl_4^{2-} and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic.
(C) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and NiCl_4^{2-} is paramagnetic.
(D) $\text{Ni}(\text{CO})_4$ is diamagnetic and NiCl_4^{2-} and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic.
52. In which of the following coordination entities, the magnitude of Δ_0 [CFSE in octahedral field] will be maximum?
(A) $[\text{Co}(\text{CN})_6]^{3-}$ (B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (C) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+}$
53. An ion M^{2+} , forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, match the complex with the appropriate colour.
(A) Green, blue and red (B) Blue, red and green
(C) Green, red and blue (D) Red, blue and green
54. Among TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} the colourless species are:
(A) CoF_6^{3-} and NiCl_4^{2-} (B) TiF_6^{2-} and CoF_6^{3-}
(C) NiCl_4^{2-} and Cu_2Cl_2 (D) TiF_6^{2-} and Cu_2Cl_2



55. In the formula of brown ring complex $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$, the magnetic moment is 3.87 B.M. The oxidation state of Fe and number of unpaired electrons present respectively are:
- (A) 1+, 3 (B) 2+, 3 (C) 3+, 3 (D) 3+, 5

Miscellaneous

56. Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect?
- (A) Chlorophyll is a green pigment in plants and contains calcium
(B) Haemoglobin is the red pigment of blood and contains iron
(C) Cyanocobalamin is B_{12} and contains cobalt
(D) Carboxypeptidase A is an enzyme and contains zinc
57. From the stability constant (hypothetical values), given below, predict which is the strongest ligand:
- (A) $\text{Cu}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}$, $K = 4.5 \times 10^{11}$
(B) $\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^{2-}$, $K = 2.0 \times 10^{27}$
(C) $\text{Cu}^{2+} + 2\text{en} \rightleftharpoons [\text{Cu}(\text{en})_2]^{2+}$, $K = 3.0 \times 10^{15}$
(D) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \rightleftharpoons [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$, $K = 9.5 \times 10^8$
58. On treatment of $[\text{Ni}(\text{NH}_3)_4]^{2+}$ with concentrated HCl, two compounds I and II having the same formula, $[\text{NiCl}_2(\text{NH}_3)_2]$ are obtained, I can be converted into II by boiling with dilute HCl. A solution of I reacts with oxalic acid to form $[\text{Ni}(\text{C}_2\text{O}_4)(\text{NH}_3)_2]$ whereas II does not react. Point out the correct statement of the following
- (A) I cis, II trans; both tetrahedral (B) I cis, II trans; both square planar
(C) I trans, II cis; both tetrahedral (D) I trans, II cis; both square planar
59. Which one of the following statement is incorrect?
- (A) Greater the formation constant (K_f) of a complex ion, greater is its stability.
(B) Greater the positive charge on the central metal ion, greater is the stability of the complex
(C) Greater is the basic character of the ligand, lesser is the stability of the complex.
(D) Chelate complexes have high stability constants.
60. The oxidation state of iron in $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$ is
- (A) 1+ (B) 2+ (C) 3+ (D) zero



ANSWER KEY

1.	(B)	2.	(B)	3.	(B)	4.	(A)	5.	(C)	6.	(B)	7.	(C)
8.	(C)	9.	(D)	10.	(C)	11.	(A)	12.	(D)	13.	(D)	14.	(B)
15.	(D)	16.	(C)	17.	(C)	18.	(D)	19.	(C)	20.	(B)	21.	(C)
22.	(B)	23.	(D)	24.	(C)	25.	(D)	26.	(C)	27.	(C)	28.	(A)
29.	(D)	30.	(B)	31.	(B)	32.	(B)	33.	(C)	34.	(C)	35.	(D)
36.	(D)	37.	(C)	38.	(C)	39.	(C)	40.	(D)	41.	(B)	42.	(A)
43.	(A)	44.	(C)	45.	(C)	46.	(A)	47.	(D)	48.	(D)	49.	(A)
50.	(A)	51.	(C)	52.	(A)	53.	(B)	54.	(D)	55.	(A)	56.	(A)
57.	(B)	58.	(B)	59.	(C)	60.	(B)						



Multiple Correct Type Question

- Which of the following exhibit geometrical isomerism (M stands for a metal, and a and b are achiral ligands)?
 (A) $\text{Ma}_2\text{b}_2(\text{Sq. Pl.})$ (B) Ma_4b_2 (C) Ma_5b (D) Ma_6
- Which of the following statement(s) are incorrect?
 (A) Those additional compounds which lose their identity in solution are called double salts.
 (B) In $\text{K}_3[\text{Fe}(\text{CN})_6]$ Fe^{2+} and CN^- ion can give quantitative identification test.
 (C) $[\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}]$ is a coordination compound.
 (D) All acids are lewis acids and σ donors.
- Which of the following statement(s) is (are) correct?
 (A) The oxidation state of iron in sodium nitroprusside $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ is +2.
 (B) $[\text{Ag}(\text{NH}_3)_2]^+$ is linear in shape.
 (C) In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridized.
 (D) In $\text{Ni}(\text{CO})_4$, the oxidation state of Ni is zero.
- Which of the following compound(s) show(s) optical isomerism.
 (A) $[\text{Pt}(\text{bn})_2]^{2+}$ (B) $[\text{CrCl}_2(\text{en})_2]^+$ (C) $[\text{Co}(\text{en})_3][\text{CoF}_6]$ (D) $[\text{Zn}(\text{gly})_2]$
- Select the correct statements ;
 (A) Potassium ferrocyanide and potassium ferricyanide can be differentiated by measuring the solid state magnetic moment.
 (B) The complex $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ can be differentiated by adding aqueous solution of barium chloride
 (C) The complex $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{Cl}$ can be differentiated by adding aqueous solution of silver nitrate.
 (D) the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ can be differentiated by measuring molar conductance
- Choose incorrect statement(s) regarding following complex ion.
 $[\text{Fe}(\text{ox})_3]^{3-}$ and $[\text{Fe}(\text{NO}_2)_6]^{3-}$
 (A) $[\text{Fe}(\text{ox})_3]^{3-}$ complex ion is more stable than $[\text{Fe}(\text{NO}_2)_6]^{3-}$.
 (B) Both complex ions are optically inactive.
 (C) Both follow Sidgwick's rule of E.A.N.
 (D) Both are paramagnetic.



7. Consider the following statements:
S₁: $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ and $[\text{CoF}_6]^{3-}$ are paramagnetic having four, five and four unpaired electrons respectively.
S₂: Low value of formation constant of a complex indicates its high thermodynamic stability.
S₃: The crystal field splitting Δ_o , depends upon the field produced by the ligand and charge on the metal ion.
and arrange in the order of true/ false.
(A) T T T (B) T F T (C) F T F (D) T F F
8. Which of the following d^n configuration of octahedral complex(es), cannot exist in both high spin and low spin forms.
(A) d^3 (B) d^5 (C) d^6 (D) d^8
9. Select incorrect statement(s) for $[\text{Cu}(\text{CN})_4]^{3-}$, $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex ion.
(A) Both $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have square planar geometry
(B) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have equal no. of unpaired electron
(C) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cd}(\text{CN})_4]^{2-}$ can be separated from the mixture on passing H_2S gas.
(D) All the three complexes have magnetic moment equal to zero.
10. Which of the following statement(s) is/are correct with respect to the crystal field theory?
(A) It considers only the metal ion d-orbitals and gives no consideration at all to other metal orbitals.
(B) It cannot account for the π bonding in complexes.
(C) The ligands are point charges which are either ions or neutral molecules
(D) The magnetic properties can be explained in terms of splitting of d- orbitals in different crystal field.
11. Which of the following statement(s) is/are incorrect?
(A) The state of hybridisation of central atom of anionic part of solid PBr_5 is sp^3d^2 .
(B) $[\text{Co}(\text{NH}_3)_6]^{2+}$ ion is easily oxidisable while $[\text{Co}(\text{NO}_2)_6]^{4-}$ is not.
(C) Bis(glycinato)zinc(II) is optically active.
(D) d_{z^2} orbital of central metal atom / ion is used in dsp^2 hybridisation.
12. 'Spin only' magnetic moment of Ni in $[\text{Ni}(\text{dmg})_2]$ is same as that found in:
(A) Ni in $[\text{NiCl}_2(\text{PPh}_3)_2]$ (B) Mn in $[\text{MnO}_4]^-$
(C) Co in $[\text{CoBr}_4]^{2-}$ (D) Pt in $[\text{Pt}(\text{H}_2\text{O})_2\text{Br}_2]$
13. Which of the following will have two stereoisomeric forms?
(A) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (B) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (C) $[\text{CoCl}_2(\text{en})_2]^+$ (D) $[\text{CoBrCl}(\text{Ox})_2]^{3-}$



14. Which complex of the following pairs has the larger value of Δ_0 ?
- (i) $[\text{Co}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ (ii) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Rh}(\text{H}_2\text{O})_6]^{3+}$
(iii) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (iv) $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$
- Select the correct one
- (A) $[\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (B) $[\text{Co}(\text{H}_2\text{O})_6]^{2+} < [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
(C) $[\text{Co}(\text{H}_2\text{O})_6]^{3+} > [\text{Rh}(\text{H}_2\text{O})_6]^{3+}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+} < [\text{CoF}_6]^{3-}$
15. Which is / are not correctly matched.
- | Complex compounds | IUPAC name |
|---|---|
| (A) $\text{K}[\text{CrF}_4\text{O}]$ | Potassium tetrafluoridooxidochromate(V) |
| (B) $\text{Na}[\text{BH}(\text{OCH}_3)_3]$ | Sodium hydridotrimethoxyborate(III) |
| (C) $[\text{Be}(\text{CH}_3\text{--CO--CH}_2\text{--CO--C}_6\text{H}_5)_2]^\circ$ | Bis(benzoylacetonato)beryllium(III) |
| (D) $\text{H}[\text{AuCl}_4]$ | Hydrogen tetrachloroaurate(III) |
16. Which of the following isomerisms is/are shown by the complex $[\text{CoCl}_2(\text{OH}_2)_2(\text{NH}_3)_2]\text{Br}$?
- (A) Ionization (B) Linkage (C) Geometrical (D) optical
17. Which of the following statement(s) is/are incorrect
- (A) In $[\text{CoBrCl}(\text{en})_2]^+$ geometrical isomerism exists, while optical isomerism does not exist
(B) Potassium aquadicyanidosuperoxidoperoxidochromate(III) is IUPAC name for $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O}_2)(\text{H}_2\text{O})]$
(C) There are 3 geometrical isomers and 15 stereoisomers possible for $[\text{Pt}(\text{NO}_2)(\text{NH}_3)(\text{NH}_2\text{OH})(\text{py})]^+$ and $[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)(\text{py})]$ respectively
(D) cis and trans forms are not diastereomers to each other
18. Three arrangements are shown for the complex $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$. Pick up the wrong statement.
- (I)

(II)

(III)
- (A) I and II are geometrical isomers (B) II and III are optical isomers
(C) I and III are optical isomers (D) II and III are geometrical isomers
19. Which of the following statement is true about the complex $[\text{CrCl}_3(\text{OH})_2(\text{NH}_3)]^{2-}$ ion.
- (A) It has three geometrical isomers.
(B) Only one space isomer is optically active and remaining are inactive.
(C) There are total four space isomers.
(D) The magnetic moment of complex ion is 3.89 B.M.



20. Consider the following complexes $[\text{V}(\text{CO})_6]^-$, $[\text{Cr}(\text{CO})_6]$ and $[\text{Mn}(\text{CO})_6]^+$. Then incorrect statement (s) about metal carbonyls is /are.
- (A) 'C–O' bond is strongest in the cation and weakest in the anion
(B) 'C–O' bond order is less in the cation than in anion.
(C) 'C–O' bond longer in the cation than in anion or neutral carbonyl.
(D) 'M–C' bond order is higher in the cation than in anionic or neutral carbonyl.
21. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?
- (A) Element always forms colourless compound.
(B) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals.
(C) It can have either d^3 or d^8 configuration.
(D) It can have either d^7 or d^8 configuration.
22. Which of the following is correct about?
Tetraamminedithiocyanato-Scobalt(III) tris(oxalato)cobaltate(III)
- (A) formula of the complex is $[\text{Co}(\text{SCN})_2(\text{NH}_3)_4][\text{Co}(\text{ox})_3]$
(B) It is a chelating complex and show linkage isomerism.
(C) It shows optical isomerism.
(D) It shows geometrical isomerism.
23. Which is correct statement(s)?
- (A) $[\text{Ag}(\text{NH}_3)_2]^+$ is linear with sp hybridised Ag^+ ion
(B) NiCl_4^{2-} , VO_4^{3-} and MnO_4^- have tetrahedral geometry
(C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$ & $[\text{Ni}(\text{CN})_4]^{2-}$ have dsp^2 hybridisation of the metal ion
(D) $\text{Fe}(\text{CO})_5$ has trigonal bipyramidal structure with $d_{z^2}sp^3$ hybridised iron.

Comprehension Type Question

Comprehension (Q.24 to Q.26)

Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.

24. For a given metal M^{3+} coordination number is six, then for which set of ligands, complex will be more stable-
- (A) $6\text{H}_2\text{O}$ (B) 6F^- (C) EDTA^{4-} (D) $2\text{H}_2\text{O}$ and $2\text{C}_2\text{O}_4^{2-}$
25. $[\text{Mn}(\text{CO})_5]$ can attain more stability by:
- (A) Oxidation of itself (B) Reduction of itself
(C) Dimerization (D) Both (B) and (C)



26. The metal cation that has least tendency to accept electron pair from NH_3 is
(A) Fe^{3+} (B) Rh^{3+} (C) Zn^{2+} (D) Ba^{++}

Comprehension (Q.27 to Q.29)

$\text{Co}^{2+} (\text{aq.}) + \text{SCN}^- (\text{aq.}) \longrightarrow \text{Complex (X)}.$

$\text{Ni}^{2+} (\text{aq.}) + \text{Dimethylglyoxime} \xrightarrow{\text{NH}_4\text{OH}} \text{Complex (Y)}.$

The coordination number of cobalt and nickel in complexes X and Y are four.

27. The IUPAC names of the complexes (X) and (Y) are respectively:
(A) tetrathiocyanato-S-cobalt(II) and bis(dimethylglyoximate) nickel(II).
(B) tetrathiocyanato-S-cobaltate (II) and bis(dimethylglyoximate)nickel (II).
(C) tetrathiocyanato-S-cobaltate (II) and bis(dimethylglyoximate)nickelate(II).
(D) tetrathiocyanato-S-cobaltate(III) and bis(dimethylglyoximate)nickel(II).
28. The geometry of complexes (X) and (Y) are respectively:
(A) tetrahedral and square planar. (B) both tetrahedral.
(C) square planar and tetrahedral (D) both square planar.
29. Select the correct statement for the complexes (X) and (Y).
(A) (X) is paramagnetic with two unpaired electrons.
(B) (Y) is diamagnetic and shows intermolecular H-bonding.
(C) (X) is paramagnetic with three unpaired electrons and (Y) is diamagnetic.
(D) (X) and (Y) both are diamagnetic.

Comprehension (Q.30 to Q.32)

The crystal field theory (C.F.T.) is now much more widely accepted than the valence bond theory. It assumes that the attraction between the central metal and the ligands in a complex is purely electrostatic. According to C.F.T. ligands are treated as a point charge and crystal field splitting energy (CFSE) increases the thermodynamic stability of the complexes. Value of CFSE depends upon nature of ligand and a spectrochemical series has been made experimentally. For tetrahedral complexes, Δ is about $4/9$ times to Δ_0 (CFSE for octahedral complexes). This energy lies in the visible region and i.e. why electronic transitions are responsible for colour.

30. Which of the following statements is not correct about C.F.T. -
(A) Diamagnetic metal ions cannot have an odd number of electrons.
(B) In an octahedral crystal field, the d electrons of a metal ion occupy the e_g set of orbitals before they occupy the t_{2g} set of orbitals.
(C) Low spin complex may be paramagnetic.
(D) In high spin octahedral complex, Δ_0 is less than the electron pairing energy and is relatively very small.



31. Crystal field stabilization energy for $[\text{CoF}_6]^{3-}$ is
 (A) $0.6 + P$ (B) $-0.4 + P$ (C) $1.2 + 2P$ (D) $2.4 + 4P$
 [P is pairing energy]
32. $\text{Ti}^{3+}(\text{aq})$ is violet while $\text{Ti}^{4+}(\text{aq})$ is colourless because -
 (A) There is no crystal field effect in Ti^{4+}
 (B) There energy difference between t_{2g} and e_g of Ti^{4+} is quite high and does not fall in the visible region.
 (C) Ti^{4+} has d^0 configuration.
 (D) Ti^{4+} is very small in comparison to Ti^{3+} and hence does not absorb any radiation.

Matrix Match Type Question

33. Match the complexes in column I with their stereo properties is column II

Column I	Column II
(A) $[\text{CoCl}_3(\text{NH}_3)_3]$	(P) Show facial isomer
(B) $[\text{Cr}(\text{OX})_3]^{3-}$	(Q) Cis form is optically active
(C) $[\text{CrCl}_2(\text{OX})_2]$	(R) Trans form is optically inactive
(D) $[\text{RhCl}_3(\text{Py})_3]$	(S) Show meridional form
	(T) Two optically active isomer

34. Match the complexes in Column-I with their stereo properties in Column II
- | Column-I | Column II |
|--|----------------------------------|
| (A) $[\text{Ma}_2\text{bcde}]^{n\pm}$ | (P) 3 optically inactive isomers |
| (B) $[\text{Ma}_2\text{b}_2\text{c}_2]^{n\pm}$ | (Q) 4 geometrical isomers |
| (C) $[\text{Ma}_3\text{bcd}]^{n\pm}$ | (R) 6 stereo(space)isomers |
| (D) $[\text{M}(\text{AB})\text{C}_2\text{d}_2]^{n\pm}$ | (S) 2 optically active isomers |

(where AB $\rightarrow$ Unsym. bidentate ligand having no chiral center, a,b,c,d & e $\rightarrow$ monodentate ligands)

ANSWER KEY

- | | | | | |
|--|--|------------|-----------|------------|
| 1. (AB) | 2. (BCD) | 3. (ABD) | 4. (ABCD) | 5. (ABCD) |
| 6. (ABC) | 7. (B) | 8. (ABC) | 9. (ABC) | 10. (ABCD) |
| 11. (ABD) | 12. (BD) | 13. (AB) | 14. (AB) | 15. (BCD) |
| 16. (ACD) | 17. (ABCD) | 18. (BCD) | 19. (AD) | 20. (BCD) |
| 21. (BC) | 22. (BCD) | 23. (ABCD) | 24. (C) | 25. (D) |
| 26. (D) | 27. (B) | 28. (A) | 29. (C) | |
| 30. (B) | 31. (B) | 32. (C) | | |
| 33. (A) $\rightarrow$ PS; (B) $\rightarrow$ T; (C) $\rightarrow$ QRT; (D) $\rightarrow$ PS | 34. (A) $\rightarrow$ P; (B) $\rightarrow$ RS; (C) $\rightarrow$ PQS; (D) $\rightarrow$ QR | | | |

Subjective Exercise - I

- How many total sodium ions are present in one formula unit of sodium ethane-1,2-diaminetetraacetatochromate (II) and sodium hexanitrito cobaltate (III)?
- A blue colour complex is obtained in the analysis of Fe^{+3} having formula $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 Let a = oxidation number of Iron in the coordination sphere
 b = no. of secondary valencies of central iron ion.
 c = Effective atomic number of Iron in the coordination sphere.
 Then find the value of $(c + a - 2b)$
- Coordination number of Cr in $\text{CrCl}_3 \cdot 5\text{H}_2\text{O}$ as six. The possible volumes of 1 M AgNO_3 needed to precipitate the chlorine in outer sphere in 200 ml of 0.1 M solution of the complex is/are.
 Write your answer as $V_1 + V_2 + V_3 + \dots$
- Ni^{+2} form a complex ion in water having the formula $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$. How many of the following statements are true for the complex ion?
 (i) The complex is octahedral in shape.
 (ii) The complex is diamagnetic in nature.
 (iii) Ni^{+2} has incompletely filled 3d subshell.
 (iv) Secondary valency of Ni^{+2} is 6.
 (v) All the bonds (metal-ligand) are perpendicular to each other.
 (vi) All the 3d orbitals of Ni^{+2} are degenerate
 (vii) Total spin of the complex is 1.
 (viii) The hybridisation of Ni^{+2} is d^2sp^3
 (ix) The complex is more stable than $[\text{Ni}(\text{en})_3]^{+2}$
 (x) Effective atomic number of Ni^{+2} is 36.
- Total number of paramagnetic complexes which are inner orbital complexes:
 (i) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (ii) $[\text{Co}(\text{NH}_3)_6](\text{NO}_3)_2$ (iii) $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$ (iv) $\text{K}_2[\text{PtCl}_6]$,
 (v) $[\text{V}(\text{H}_2\text{O})_6]\text{SO}_4$ (vi) $[\text{Mn}(\text{NH}_3)_6]\text{SO}_4$ (vii) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$
 (viii) $\text{K}_3[\text{CuCl}_4]$ (ix) $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$
- The number of coordination isomers possible for $[\text{Fe}(\text{NH}_3)_6]^{3+} [\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ is _____
- Find the sum of number of geometrical isomers for following complexes.
 (a) $[\text{CoCl}_2\text{Br}_2]^{2-}$ (b) $[\text{Rh}(\text{en})_3]^{3+}$ (c) $[\text{Cr}(\text{en})_2\text{Br}_2]^+$ (d) $[\text{Pt en Cl}_2]$
 (e) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
- How many isomeric forms are possible for the octahedral complex, $[\text{Rh}(\text{en})_2(\text{NO}_2)(\text{SCN})]^+$?

ANSWER KEY

- | | | | |
|----------------|---------|--------------------------------|--------------------------|
| 1. (05) | 2. (26) | 3. (40 + 20) ml = 60 ml | 4. (4) (i, iii, iv, vii) |
| 5. (2) (i & v) | 6. (4) | 7. (4) (0 + 0 + 2 + 0 + 2 = 4) | 8. (12) |



Subjective Exercise - II

Comprehension (Q.1 to Q.5)

In coordination chemistry there are a variety of methods applied to find out the structure of complexes. One method involves treating the complex with known reagents and from the nature of reaction, the formula of the complex can be predicted. An isomer of the complex $\text{Co(en)}_2(\text{H}_2\text{O})\text{Cl}_2\text{Br}$, on reaction with concentrated H_2SO_4 (dehydrating agent) it suffers loss in weight and on reaction with AgNO_3 solution it gives a white precipitate which is soluble in $\text{NH}_3(\text{aq})$.

1. What will be the correct coordination number of the complex:
2. If all the ligands in the coordination sphere of the above complex be replaced by F^- , then the magnetic moment of the complex ion (due to spin only) will be:
3. Similarly if all the ligands in the coordination sphere be replaced by NO_2^- , then the magnetic moment of the complex ion (due to spin only) will be:
4. If one mole of original complex is treated with excess $\text{Pb}(\text{NO}_3)_2$ solution, then the number of moles of white precipitate (of PbCl_2) formed will be:
5. The number of geometrical isomers of the formula of the above original complex are (including the complex):
6. How many total sodium ions are present in one formula unit of sodium ethane-1,2-diaminetetraacetatochromate(II) and sodium hexanitrito cobaltate(III) ?
7. A blue colour complex is obtained in the analysis of Fe^{+3} having formula $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
Let a = oxidation number of Iron in the coordination sphere
b = no. of secondary valencies of central iron ion.
c = Effective atomic number of Iron in the coordination sphere.
Then find the value of $(c + a - 2b)$
8. Ni^{+2} form a complex ion in water having the formula $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$. How many of the following statements are true for the complex ion ?
 - (i) The complex is octahedral in shape.
 - (ii) The complex is diamagnetic in nature.
 - (iii) Ni^{+2} has incompletely filled 3d subshell.
 - (iv) Secondary valency of Ni^{+2} is 6.
 - (v) All the bonds (metal-ligand) are perpendicular to each other.
 - (vi) All the 3d orbitals of Ni^{+2} are degenerate
 - (vii) Total spin of the complex is 1.
 - (viii) The hybridisation of Ni^{+2} is d^2sp^3
 - (ix) The complex is more stable than $[\text{Ni}(\text{en})_3]^{+2}$
 - (x) Effective atomic number of Ni^{+2} is 36.



9. Total number of paramagnetic complexes which are inner orbital complexes:
(i) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (ii) $[\text{Co}(\text{NH}_3)_6](\text{NO}_3)_2$ (iii) $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$
(iv) $\text{K}_2[\text{PtCl}_6]$ (v) $[\text{V}(\text{H}_2\text{O})_6]\text{SO}_4$ (vi) $[\text{Mn}(\text{NH}_3)_6]\text{SO}_4$
(vii) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$ (viii) $\text{K}_3[\text{CuCl}_4]$ (ix) $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$
10. The number of coordination isomers possible for $[\text{Fe}(\text{NH}_3)_6]^{3+}$ & $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ is _____
11. Find the sum of number of geometrical isomers for following complexes.
(a) $[\text{CoCl}_2\text{Br}_2]^{2-}$ (b) $[\text{Rh}(\text{en})_3]^{3+}$ (c) $[\text{Cr}(\text{en})_2\text{Br}_2]^+$ (d) $[\text{Pt}(\text{en})\text{Cl}_2]$
(e) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

ANSWER KEY

- | | | | | |
|-------------------------------|----------|------------------------|--------------|--------|
| 1. (6) | 2. (4.9) | 3. (0) | 4. (0.5) | 5. (2) |
| 6. 05 | 7. 26 | 8. 4 (i, iii, iv, vii) | 9. 2 (i & v) | 10. 4 |
| 11. 4 (0 + 0 + 2 + 0 + 2 = 4) | | | | |



JEE-Main (Previous Year Questions)

1. Which of the following complex species is not expected to exhibit optical isomerism?
(1) $[\text{Co}(\text{en})_3]^{3+}$ (2) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ [JEE(Main)-2013]
(3) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (4) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$
2. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is: [JEE(Main)-2014]
(1) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$ (2) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$
(3) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$ (4) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
3. The equation which is balanced and represents the correct product (s) is: [JEE(Main)-2014]
(1) $[\text{Mg}(\text{H}_2\text{O})_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg}(\text{EDTA})]^{2+} + 6\text{H}_2\text{O}$
(2) $\text{CuSO}_4 + 4\text{KCN} \rightarrow \text{K}_2[\text{Cu}(\text{CN})_4] + \text{K}_2\text{SO}_4$
(3) $\text{Li}_2\text{O} + 2\text{KCl} \rightarrow 2\text{LiCl} + \text{K}_2\text{O}$
(4) $[\text{CoCl}(\text{NH}_3)_5]^+ + 5\text{H}^+ \rightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$
4. Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is: [JEE(Main)-2014]
(1) CN^- (2) NH_3 (3) CO (4) F^-
5. Which one of the following complexes will most likely absorb visible light? [JEE(Main)-2014]
(At nos. Sc = 21, Ti = 22, V = 23, Zn = 30):-
(1) $[\text{Ti}(\text{NH}_3)_6]^{4+}$ (2) $[\text{V}(\text{NH}_3)_6]^{3+}$ (3) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ (4) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$
6. Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[\text{NiL}_4]^{2+}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively: [JEE(Main)-2014]
(1) sp^3 , zero (2) sp^3 , two (3) dsp^2 , one (4) dsp^2 , zero
7. The color of KMnO_4 is due to: [JEE(Main)-2015]
(1) $\text{L} \rightarrow \text{M}$ charge transfer transition (2) $\sigma - \sigma$ transition
(3) $\text{M} \rightarrow \text{L}$ charge transfer transition (4) $\text{d} - \text{d}$ transition
8. Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals? [JEE(Main)-2015]
(1) $[\text{CoF}_6]^{3-}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$ (3) $[\text{FeF}_6]^{3-}$ (4) $[\text{Co}(\text{NH}_3)_6]^{2+}$

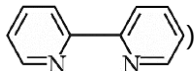


9. When concentrated HCl is added to an aqueous solution of CoCl_2 , its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction: [JEE(Main)-2015]
(1) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{CoCl}_6]^{3-}$ (3) $[\text{CoCl}_4]^{2-}$ (4) $[\text{CoCl}_6]^{4-}$
10. Identify the correct trend given below: [JEE(Main)-2016]
(Atomic No.= Ti: 22, Cr: 24 and Mo: 42)
(1) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(2) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(3) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(4) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
11. Consider the following reaction and statements: [JEE(Main)-2018]
 $[\text{Co}(\text{NH}_3)_4\text{Br}_2]^+ + \text{Br}^- \rightarrow [\text{Co}(\text{NH}_3)_3\text{Br}_3] + \text{NH}_3$
(I) Two isomers are produced if the reactant complex ion is a cis-isomer
(II) Two isomers are produced if the reactant complex ion is a trans-isomer
(III) Only one isomer is produced if the reactant complex ion is a trans-isomer.
(IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.
(1) (II) and (IV) (2) (I) and (II) (3) (I) and (III) (4) (III) and (IV)
12. The oxidation states of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{C}_6\text{H}_6)_2]$, and $\text{K}_2[\text{Cr}(\text{CN})_2(\text{O})_2(\text{O}_2)(\text{NH}_3)]$ respectively are [JEE(Main)-2018]
(1) +3, 0, and + 4 (2) +3, +4, and + 6
(3) +3, +2, and + 4 (4) +3, 0, and +6
13. Two complexes $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (A) and $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) are violet and yellow coloured, respectively. The incorrect statement regarding them is: [JEE(Main)-2019]
(1) Δ_0 value of (A) is less than that of (B).
(2) Δ_0 value of (A) and (B) are calculated from the energies of violet and yellow light, respectively
(3) Both absorb energies corresponding to their complementary colors.
(4) Both are paramagnetic with three unpaired electrons.
14. The complex that has highest crystal field splitting energy (Δ) is: [JEE(Main)-2019]
(1) $\text{K}_2[\text{CoCl}_4]$ (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
(3) $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_3$ (4) $\text{K}_3[\text{Co}(\text{CN})_6]$
15. Wilkinson catalyst is: [JEE(Main)-2019]
(1) $[(\text{Et}_3\text{P})_3\text{RhCl}]$ (Et = C_2H_5) (2) $[(\text{Ph}_3\text{P})_3\text{RhCl}]$
(3) $[(\text{Ph}_3\text{P})_3\text{IrCl}]$ (4) $[(\text{Et}_3\text{P})_3\text{IrCl}]$



16. The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is:
(1) 4 (2) 8 (3) 12 (4) 16 [JEE(Main)-2019]
17. A reaction of cobalt (III) chloride and ethylenediamine in a 1: 2 mole ratio generates two isomeric products A (violet coloured) and B (green coloured). A can show optical activity, B is optically inactive. What type of isomers does A and B represent?
(1) Coordination isomers (2) Ionisation isomers [JEE(Main)-2019]
(3) Linkage isomers (4) Geometrical isomers
18. The number of bridging CO ligand(s) and Co-Co bond(s) in $Co_2(CO)_8$, respectively are:-
(1) 0 and 2 (2) 4 and 0 (3) 2 and 1 (4) 2 and 0 [JEE(Main)-2019]
19. The metal d-orbitals that are directly facing the ligands in $K_3[Co(CN)_6]$ are:
(1) d_{xy} , d_{xz} and d_{yz} (2) $d_{x^2-y^2}$ and d_{z^2} (3) d_{xy} and $d_{x^2-y^2}$ (4) d_{xz} , d_{yz} and d_{z^2} [JEE(Main)-2019]
20. The theory that can completely/properly explain the nature of bonding in $[Ni(CO)_4]$ is:
(1) Werner's theory (2) Valence bond theory [JEE(Main)-2020]
(3) Molecular orbital theory (4) Crystal field theory
21. Among the statements (a)-(d), the incorrect ones are: [JEE(Main)-2020]
(a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments
(b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{eg}^4 e_g^2$
(c) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3-}$
(d) If the Δ_0 for an octahedral complex of Co(III) is $18,000\text{ cm}^{-1}$, the Δ_t for its tetrahedral complex with the same ligand will be $16,000\text{ cm}^{-1}$
(1) (a) and (d) only (2) (a) and (b) only (3) (c) and (d) only (4) (b) and (c) only
22. $[Pd(F)(Cl)(Br)(I)]^{2-}$ has an number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[Fe(CN)_6]^{n-6}$, respectively are:
[Note: Ignore the pairing energy] [JEE(Main)-2020]
(1) 1.73 BM and $-2.0\Delta_0$ (2) 2.84 BM and $-1.6\Delta_0$
(3) 5.92 BM and 0 (4) 0 BM and $-2.4\Delta_0$



23. Given below are two statements: [JEE(Main)-2021]
Statement I: The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH
Statement II: The dimethyl glyoxime is a bidentate neutral ligand.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and statement II are true
(2) Both statement I and statement II are false
(3) Statement I is false but statement II is true
(4) Statement I is true but statement II is false
24. Reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g} orbitals of the product is _____. [JEE(Main)-2022]
25. The correct order of energy of absorption for the following metal complexes is
A: $[\text{Ni}(\text{en})_3]^{2+}$, B: $[\text{Ni}(\text{NH}_3)_6]^{2+}$, C: $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ [JEE(Main)-2022]
(1) $\text{C} < \text{B} < \text{A}$ (2) $\text{B} < \text{C} < \text{A}$ (3) $\text{C} < \text{A} < \text{B}$ (4) $\text{A} < \text{C} < \text{B}$
26. Low oxidation state of metals in their complexes are common when ligands:
(1) have good π -accepting character [JEE(Main)-2022]
(2) have good σ -donor character
(3) are having good π -donating ability
(4) are having poor σ -donating ability
27. Total number of relatively more stable isomer(s) possible for octahedral complex $[\text{Cu}(\text{en})_2(\text{SCN})_2]$ will be _____. [JEE(Main)-2022]
28. Sum of oxidation state (magnitude) and coordination number of cobalt in $\text{Na}[\text{Co}(\text{bpy})\text{Cl}_4]$ is _____. [JEE(Main)-2022]
(Given $\text{bpy} =$ )

ANSWER KEY

- | | | | | | | | | | | | | | |
|-----|-----|-----|-----|-----|------|-----|-----|-----|-----|-----|------|-----|------|
| 1. | (3) | 2. | (4) | 3. | (4) | 4. | (3) | 5. | (2) | 6. | (4) | 7. | (1) |
| 8. | (3) | 9. | (3) | 10. | (2) | 11. | (3) | 12. | (4) | 13. | (2) | 14. | (4) |
| 15. | (2) | 16. | (3) | 17. | (4) | 18. | (3) | 19. | (2) | 20. | (3) | 21. | (1) |
| 22. | (1) | 23. | (4) | 24. | (06) | 25. | (1) | 26. | (1) | 27. | (03) | 28. | (09) |



JEE-Advanced (Previous Year Questions)

1. The complex ion which has no 'd' electrons in the central metal atom is:
[Atomic number Cr = 24, Mn = 25, Fe = 26, Co = 27] [JEE-2001]
(A) $[\text{MnO}_4]^-$ (B) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
2. The correct order of hybridisation of the central atom in the following species.
 NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is [At No. Pt = 78] [JEE-2001]
(A) $\text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$ and sp^3 (B) $\text{sp}^3, \text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$
(C) $\text{dsp}^2, \text{sp}^2, \text{sp}^3$ and sp^3d (D) $\text{dsp}^2, \text{sp}^3, \text{sp}^2$ and sp^3d
3. The species having tetrahedral shape is: [JEE-2004]
(A) $[\text{PdCl}_4]^{2-}$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$ (C) $[\text{Pd}(\text{CN})_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
4. The pair of compounds having metals in their highest oxidation state is [JEE-2004]
(A) MnO_2 , FeCl_3 (B) $[\text{MnO}_4]^-$, CrO_2Cl_2
(C) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_6]^{-3}$ (D) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
5. Spin only magnetic moment of the compound $\text{Hg}^{+2}[\text{Co}(\text{SCN}_4)]$ is [JEE-2004]
(A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$
6. Which of the following pair is expected to exhibit same colour in solution? [JEE-2005]
(A) VOCl_2 ; FeCl_2 (B) CuCl_2 ; VOCl_2 (C) MnCl_2 ; FeCl_2 (D) FeCl_2 ; CuCl_2
7. Which type of isomerism is shown by $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{Cl}$? [JEE-2005]
(A) Geometrical and Ionisation (B) Optical and Ionisation
(C) Geometrical and Optical (D) Geometrical only

Paragraph Question No. 8 to 10 (3 questions)

The coordination number of Ni^{2+} is 4.

$\text{NiCl}_2 + \text{KCN (excess)} \rightarrow \text{A (cyano complex)}$

$\text{NiCl}_2 + \text{KCl (excess)} \rightarrow \text{B (chloro complex)}$

8. The IUPAC name of A and B are [JEE-2006]
(A) Potassium tetracyanonickelate (II), potassium tetrachloridonickelate (II)
(B) Tetracyanidopotassiumnickelate (II), tetrachloridopotassiumnickelate (II)
(C) Tetracyanidonickel (II), tetrachloridonickel (II)
(D) Potassium tetracyanonickel (II), potassium tetrachloridonickel (II)



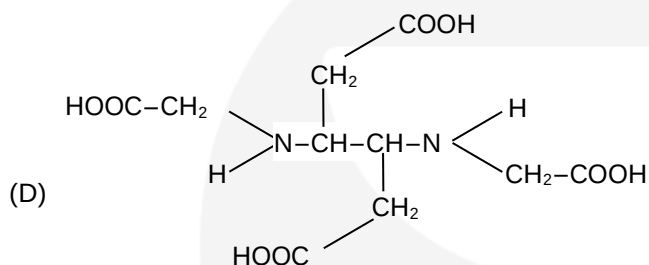
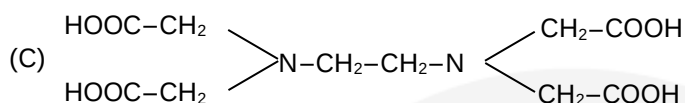
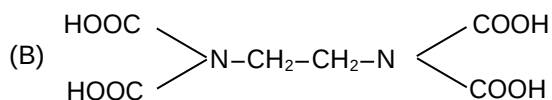
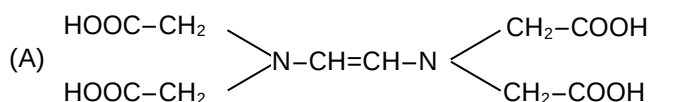
9. Predict the magnetic nature of A and B. [JEE-2006]
(A) Both are diamagnetic.
(B) A is diamagnetic and B is paramagnetic with one unpaired electron.
(C) A is diamagnetic and B is paramagnetic with two unpaired electrons.
(D) Both are paramagnetic.
10. The hybridization of A and B are [JEE-2006]
(A) dsp^2 , sp^3 (B) sp^3 , sp^3 (C) dsp^2 , dsp^2 (D) sp^3d^2 , d^2sp^3
11. If the bond length of CO bond in carbon monoxide is $1.128\text{\AA}$, then what is the value of CO bond length in $\text{Fe}(\text{CO})_5$? [JEE-2006]
(A) $1.15\text{\AA}$ (B) $1.128\text{\AA}$ (C) $1.72\text{\AA}$ (D) $1.118\text{\AA}$
12. Among the following metal carbonyls, the C–O bond order is lowest in [JEE-2007]
(A) $[\text{Mn}(\text{CO})_6]^+$ (B) $[\text{Fe}(\text{CO})_5]$ (C) $[\text{Cr}(\text{CO})_6]$ (D) $[\text{V}(\text{CO})_6]^-$
13. Match the complexes in Column I with their properties listed in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE-2007]
- | Column I | Column II |
|---|---------------------------------------|
| (A) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$ | (P) Geometrical isomers |
| (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ | (Q) Paramagnetic |
| (C) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}$ | (R) Diamagnetic |
| (D) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ | (S) Metal ion with 2+ oxidation state |
14. Among the following, the coloured compound is [JEE-2008]
(A) CuCl (B) $\text{K}_3[\text{Cu}(\text{CN})_4]$
(C) CuF_2 (D) $[\text{Cu}(\text{CH}_3\text{CN})_4] \text{BF}_4$
15. The IUPAC name of $[\text{Ni}(\text{NH}_3)_4]^{+2}[\text{NiCl}_4]^{-2}$ is [JEE-2008]
(A) Tetrachloronickel (II)-tetraamminenickel (II)
(B) Tetraamminenickel (II)-tetrachloronickel (II)
(C) Tetraamminenickel (II)-tetrachloronickelate (II)
(D) Tetrachloronickel (II)-tetraamminenickelate (0)
16. Both $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are [JEE-2008]
(A) sp^3 , sp^3 (B) sp^3 , dsp^2 (C) dsp^2 , sp^3 (D) dsp^2 , dsp^2



17. Statement-1: The geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ are optically inactive.
Statement-2: Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ possess axis of symmetry.
(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True [JEE-2008]
18. Statement-1: $[Fe(H_2O)_5NO]SO_4$ is paramagnetic [JEE-2008]
Statement-2: The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True
19. The spin only magnetic moment value (in Bohr magneton units) of $Cr(CO)_6$ is [JEE-2009]
(A) 0 (B) 2.84 (C) 4.90 (D) 5.92
20. The compound(s) that exhibit(s) geometrical isomerism is (are): [JEE-2009]
(A) $[Pt(en)Cl_2]$ (B) $[Pt(en)_2]Cl_2$
(C) $[Pt(en)_2Cl_2]Cl_2$ (D) $[Pt(NH_3)_2Cl_2]$
21. The number of water molecule(s) directly bonded to the metal centre in $CuSO_4 \cdot 5H_2O$ is. [JEE-2009]
22. The ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is – [JEE-2010]
(A) $[Cr(H_2O)_4(O_2N)]Cl_2$ (B) $[Cr(H_2O)_4Cl_2](NO_2)$
(C) $[Cr(H_2O)_4Cl(ONO)]Cl$ (D) $[Cr(H_2O)_4Cl_2(NO_2)] \cdot H_2O$
23. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is. [JEE-2010]



24. The correct structure of ethylenediaminetetraacetic acid (EDTA) is – [JEE-2010]



25. Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl^- , CN^- and H_2O respectively, are – [JEE-2011]

- (A) octahedral, tetrahedral and square planar
(B) tetrahedral, square planar and octahedral
(C) square planar, tetrahedral and octahedral
(D) octahedral, square planar and octahedral

26. Among the following complexes (K–P) [JEE-2011]

$\text{K}_3[\text{Fe}(\text{CN})_6]$ (K), $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (L), $\text{Na}_3[\text{Co}(\text{ox})_3]$ (M), $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ (N),
 $\text{K}_2[\text{Pt}(\text{CN})_4]$ (O) and $[\text{Zn}(\text{H}_2\text{O})_6](\text{NO}_3)_2$ (P)

The diamagnetic complex are –

- (A) K, L, M, N (B) K, M, O, P (C) L, M, O, P (D) L, M, N, O

27. As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is:

- (A) Tetraaquadiaminecobalt(III) chloride
(B) Tetraaquadiamminecobalt(III) chloride
(C) Diaminetetraaquacobalt(III) chloride
(D) Diamminetetraaquacobalt(III) chloride

[JEE-2012]



28. The colour of light absorbed by an aqueous solution of CuSO_4 is - [JEE-2012]
(A) orange-red (B) blue-green (C) yellow (D) violet
29. $\text{NiCl}_2\{\text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)\}_2$ exhibits temperature dependent magnetic behaviour (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively: [JEE-2012]
(A) tetrahedral and tetrahedral (B) square planar and square planar
(C) tetrahedral and square planar (D) square planar and tetrahedral
30. Consider the following complex ions P, Q and R ,
 $\text{P} = [\text{FeF}_6]^{3-}$, $\text{Q} = [\text{V}(\text{H}_2\text{O})_6]^{2+}$ and $\text{R} = [\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is - [JEE-2013]
(A) $\text{R} < \text{Q} < \text{P}$ (B) $\text{Q} < \text{R} < \text{P}$ (C) $\text{R} < \text{P} < \text{Q}$ (D) $\text{Q} < \text{P} < \text{R}$
31. EDTA^{4-} is ethylenediaminetetraacetate ion. The total number of N–Co–O bond angles in $[\text{Co}(\text{EDTA})]^{-1}$ complex ion is [JEE-2013]
32. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) – [JEE-2013]
(A) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ and $[\text{Pt}(\text{NH}_3)_2(\text{H}_2\text{O})\text{Cl}]^+$
(C) $[\text{CoBr}_2\text{Cl}_2]^{2-}$ and $[\text{PtBr}_2\text{Cl}_2]^{2-}$ (D) $[\text{Pt}(\text{NH}_3)_3(\text{NO}_3)] \text{Cl}$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}] \text{Br}$
33. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. [JEE(Advanced)-2014]

{en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ ' atomic numbers; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I

List-II

- | | | | |
|-----|---|-----|--|
| (P) | $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ | (1) | Paramagnetic and exhibits ionisation isomerism |
| (Q) | $[\text{Ti}(\text{H}_2\text{O})_5\text{Cl}](\text{NO}_3)_2$ | (2) | Diamagnetic and exhibits cis-trans isomerism |
| (R) | $[\text{Pt}(\text{en})(\text{NH}_3)\text{Cl}]\text{NO}_3$ | (3) | Paramagnetic and exhibits cis-trans isomerism |
| (S) | $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{NO}_3$ | (4) | Diamagnetic and exhibits ionisation isomerism |

Code:

- | | (P) | (Q) | (R) | (S) |
|-----|-----|-----|-----|-----|
| (A) | 4 | 2 | 3 | 1 |
| (D) | 1 | 3 | 4 | 2 |
| (B) | 3 | 1 | 4 | 2 |
| (D) | 1 | 3 | 4 | 2 |



34. A list of species having the formula XZ_4 is given below: [JEE(Advanced)-2014]
 XeF_4 , SF_4 , SiF_4 , BF_4^- , BrF_4^- , $[Cu(NH_3)_4]^{2+}$, $[FeCl_4]^{2-}$, $[CoCl_4]^{2-}$ and $[PtCl_4]^{2-}$.
Defining shape on the basis of the location of X and Z atoms, the total number of species having a square planar shape is
35. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbitals of the transition metal in each case. [JEE(Advanced)-2000]
36. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridisation of chromium in each. Calculate their magnetic moments (spin-only value). [JEE(Advanced)-2001]
37. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridisation of the metal ion. Calculate the magnetic moment (spin only) of the species. [JEE(Advanced)-2002]
38. Write the IUPAC name of the compound $K_2[Cr(NO)(CN)_4(NH_3)]$. Spin magnetic moment of the complex $\mu = 1.73$ BM. Give the structure of anion. [JEE(Advanced)-2003]
39. $NiCl_2$ in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH , giving a bright red colour. [JEE(Advanced)-2004]
(a) Draw its structure and show H-bonding
(b) Give oxidation state of Ni and its hybridisation
(c) Predict whether it is paramagnetic or diamagnetic
40. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is: [Atomic number of Fe = 26] [JEE(Advanced)-2015]
41. In the complex acetyl bromidodicarbonylbis (triethylphosphine)iron(II), the number of Fe–C bond(s) is- [[JEE(Advanced)-2015]
42. Among the complex ions, $[Co(NH_2-CH_2-CH_2-NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2-CH_2-CH_2-NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) cis-trans isomerism is – [JEE(Advanced)-2015]



43. The geometries of the ammonia complexes of Ni^{2+} , Pt^{2+} and Zn^{2+} , respectively, are:
(A) octahedral, square planar and tetrahedral [JEE(Advanced)-2016]
(B) square planar, octahedral and tetrahedral
(C) tetrahedral, square planar and octahedral
(D) octahedral, tetrahedral and square planar
44. Addition of excess aqueous ammonia to a pink coloured aqueous solute ion of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1: 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. [JEE(Advanced)-2017]
Among the following options, which statement(s) is(are) correct?
(A) The hybridization of the central metal ion in Y is $d^2 sp^3$
(B) Z is a tetrahedral complex
(C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
(D) When X and Z are in equilibrium at 0°C , the colour of the solution is pink
45. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are) (Atomic numbers: Fe = 26, Ni = 28) [JEE(Advanced)-2018]
(A) Total number of valence shell electrons at metal centre in $\text{Fe}(\text{CO})_5$ or $\text{Ni}(\text{CO})_4$ is 16
(B) These are predominantly low spin in nature
(C) Metal-carbon bond strengthens when the oxidation state of the metal is lowered
(D) The carbonyl C-O bond weakens when the oxidation state of the metal is increased
46. The correct option(s) regarding the complex $[\text{Co}(\text{en})(\text{NH}_3)_3(\text{H}_2\text{O})]^{3+}$ (en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) is (are) [JEE(Advanced)-2018]
(A) It has two geometrical isomers
(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
(C) It is paramagnetic
(D) It absorbs light at longer wavelength as compared to $[\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$
47. Total number of cis N-Mn-Cl bond angles (that is, Mn-N and Mn-Cl bonds in cis positions) present in a molecule of cis- $[\text{Mn}(\text{en})_2\text{Cl}_2]$ complex is ____ (en = $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) [JEE(Advanced)-2019]
48. Choose the correct statement(s) among the following: [JEE(Advanced)-2020]
(A) $[\text{FeCl}_4]^-$ has tetrahedral geometry.
(B) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has 2 geometrical isomers.
(C) $[\text{FeCl}_4]^-$ has higher spin-only magnetic moment than $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$.
(D) The cobalt ion in $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has sp^3d^2 hybridization.



49. The total number of possible isomers for $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ is _____. [JEE(Advanced)-2021]

50. LIST-I contains metal species and LIST-II contains their properties. [JEE(Advanced)-2022]

LIST-I	LIST-II
(I) $[\text{Cr}(\text{CN})_6]^{4-}$	(P) t_{2g} orbitals contain 4 electrons
(II) $[\text{RuCl}_6]^{2-}$	(Q) $\mu(\text{spin-only}) = 4.9 \text{ BM}$
(III) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$	(R) low spin complex ion
(IV) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(S) metal ion in 4+ oxidation state
	(T) d^4 species

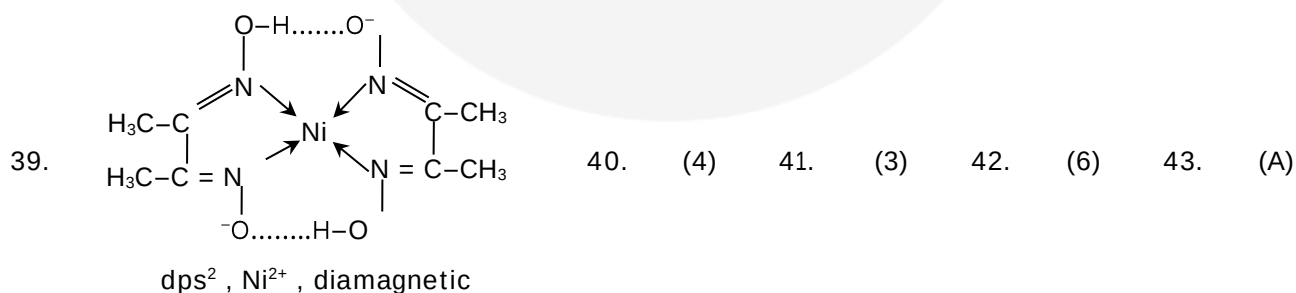
[Given: Atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option

- (A) I $\rightarrow$ R, T; II $\rightarrow$ P, S; III $\rightarrow$ Q, T; IV $\rightarrow$ P, Q (B) I $\rightarrow$ R, S; II $\rightarrow$ P, T; III $\rightarrow$ P, Q; IV $\rightarrow$ Q, T
 (C) I $\rightarrow$ P, R; II $\rightarrow$ R, S; III $\rightarrow$ R, T; IV $\rightarrow$ P, T (D) I $\rightarrow$ Q, T; II $\rightarrow$ S, T; III $\rightarrow$ P, T; IV $\rightarrow$ Q, R

ANSWER KEY

1. (A) 2. (B) 3. (D) 4. (B) 5. (B) 6. (B) 7. (A)
 8. (A) 9. (C) 10. (A) 11. (A) 12. (D)
 13. (A) $\rightarrow$ P,Q,S; (B) $\rightarrow$ P,R,S; (C) $\rightarrow$ Q,S; (D) $\rightarrow$ Q,S 14. (C) 15. (C) 16. (B)
 17. (B) 18. (A) 19. (A) 20. (CD) 21. (4) 22. (B) 23. (3)
 24. (C) 25. (B) 26. (C) 27. (D) 28. (A) 29. (C) 30. (B)
 31. (8) 32. (BD) 33. (B) 34. (4) 35. d^2sp^3 , dsp^2 and sp^3
 36. $\sqrt{15} \text{ BM}$ 37. $\sqrt{8}, 0$
 38. Potassium amminetetracyanonitrosoniumchromate $\rightarrow d^2sp^3$, octahedral



44. (ABD) 45. (BC) 46. (ABD) 47. (6) 48. (AC) 49. (6)
 50. (A)